



A Comprehensive Review on Machine Learning-based Approaches for Next Generation Wireless Network

Suman Paul¹

Received: 23 June 2023 / Accepted: 25 March 2024
© The Author(s), under exclusive licence to Springer Nature Singapore Pte Ltd. 2024

Abstract

Rapid progress of wireless communication and networks plays a critical role in different applications related to carrying out heterogeneous ever-increasing traffic in multimedia communication, remote health, and commercial applications. The fastest developments and uninterrupted upgradations in recent telecommunication networks regarding the deployment of the advanced 5G, which is being rolled out, draw attention to various research challenges and demanding transition requirements from existing 4G to 5G, beyond (B5G) and toward the 6G. These network upgradations strongly stipulate the need for system flexibility along with intelligent decisions at run-time with the integration of Artificial Intelligence (AI) and Machine Learning (ML)-based techniques. In this paper, various research challenges regarding the rolling out of advanced 5G network, the need for artificial intelligence in the rapidly emerging next generation wireless network (NGWN) along with various state-of-the-art approaches of the Artificial Intelligence and Machine learning, target applications, challenges regarding the incorporation of intelligence in the network have been investigated in a unified way. The paper highlights the contemporary cutting-edge developments of the telecom industry, research trends of network edge intelligence including the integration of federated learning (FL), allied challenges and open research issues in promising 6G networks.

Keywords Artificial Intelligence (AI) · Machine learning (ML) · Beyond fifth generation (B5G) and sixth generation (6G) networks · Network intelligence · Next generation wireless network (NGWN) · Reinforcement learning (RL) · Deep learning (DL) · Federated learning (FL) · Mobile edge computing (MEC) · Edge intelligence (EI)

Abbreviations

AI	Artificial Intelligence
ANN	Artificial Neural network
B5G	Beyond 5G
CNN	Convolutional Neural Network
CSI	Channel State Information
CSP	Communication Service Provider
C-RAN	Cloud Radio Access Networks
D2D	Device to Device
DL	Deep Learning
DRL	Deep Reinforcement Learning
eMBB	Enhanced Mobile Broadband
FL	Federated Learning
HMM	Hidden Markov Model
IoT	Internet of things
KNN	K-Nearest Neighbour

ITU	International Telecommunication Union
M2M	Machine to Machine
MDP	Markov Decision Process
MIMO	Multiple Input Multiple Output
ML	Machine Learning
mMTC	Machine-to-Machine Communication
NGWN	Next Generation Wireless Network
NOMA	Non-Orthogonal Multiple Access
NVF	Network Functions Virtualization
QoS	Quality of service
QoE	Quality of Experience
RL	Reinforcement Learning
RNN	Recurrent Neural Network
SDN	Software-Defined Networking
SOM	Self-organizing Map
SON	Self-Organizing Network
SVM	Support Vector Machine
ToD	Tele-operated driving
UAV	Unmanned Aerial Vehicle
VLC	Visible Light Communication
URLLC	Ultra Reliable Low Latency Communication

✉ Suman Paul
paulsuman999@gmail.com

¹ Department of ECE, Haldia Institute of Technology, Haldia, West Bengal 721 657, India

Introduction

Wireless networks and communication systems have experienced significant revolutionary progress in modern days. With the fastest developments of the standardization of 5G phase 2, the various applications are rigorously being developed and advanced 5G is rolling out commercially in several countries [1]. Next-generation communication systems aim to achieve high spectral and energy efficiency. Moreover, it targets massive connectivity and low-latency communication due to extensive growth in the number of Internet of Things (IoT) devices. The aim of incorporation of artificial intelligence (AI) and machine learning (ML) based approaches in advanced 5G networks and beyond is for continuous up gradations and network performance improvement.

The three important key components that play a vital role in the performance improvement of the 5G network are (i) enhanced mobile broadband (eMBB), (ii) ultrareliable low-latency communications (uRLLC) and (iii) massive machine-to-machine communication (mMTC). The eMBB is designed to offer higher data rates, better

bandwidth, higher throughput, low latency, increased reliability, and enhanced multimedia functionality for the end users. The uRLLC has been developed to meet the requirements of Industry 4.0. It is a service that aims to deliver data packets in applications requiring improved availability, reliability, and minimized latency. The mMTC is a type of communication between the machines over wireless networks where data is generated, information exchanged and actuation occurs with least or minimal human intervention. The mMTC enables ultra-low latency connections, low energy consumption and improved network performance for an extensive variety of IoT use cases. The specific characteristics of eMBB, uRLLC and mMTC are stated in Table 1.

With the developments of 5G, several applications related to device-to-device (D2D) communications, Internet of Things (IoT), machine-to-machine (M2M) communications, Internet of Vehicles (IoV), wearable technology, and health-care have been focused for potential research and developments. The typical use cases of the 5G and its advances are [2] cloud-based virtual and augmented reality including computer-based modelling, computer gaming (real-time), autonomous driving, Tele-operated driving (ToD) [2] and

Table 1 Comparative key attributes [17, 119] and KPI [20, 21] between 5G advanced and Target 6G

Attributes	5G Advanced	6G
Latency	1 ms	< 1 ms
Peak data rates	100 Gbps	1 Tbps
Frequency band	Sub 6 GHz, mm Wave for fixed access	Sub 6G Hz mm Wave for mobile access, terahertz band, Non-RF e.g., Visible Light Communication (VLC)
Technology	Communication in D2D, in a network (ultra-dense), NOMA (Non-Orthogonal Multiple Access), access of small cells, relaying	Communication: Quantum; Visible light; Haptic technology; hybrid access
Network type (Implementation)	Network slicing, NFV, SDN (Software-Defined Networking)	Incorporate extreme Intelligence: AI/ ML based approaches (embrace deep learning) for: SDN, NFV, cloud, network slicing
Computing Techniques	Cloud computing, trends towards fog computing	AI based quantum computing, edge computing
Mobility	500 km h ⁻¹	700 km h ⁻¹
Flow Density	10 Tbit/ (s-km ²)	100–1000 Tbit/ (s-km ²)
Architecture	Small cells (Dense sub-6 GHz)	Smart surfaces (cell-free), Temporary Applicable hotspots, use of drones applicable for base stations
Frequency	3–300 GHz	1000 Ghz
Target Base station Computing Power	100–200 Tops	1000 Tops
Efficiency(Spectral)	30 bps/ Hz	100 bps/Hz
Device services	Secured connectivity	Real-time physical interactions
Reliability	99.9999%	99.99999%
Position precision	1–10 m	0.1–1 m
Coverage	Partial	Global
Various Applications	(i)Reliable eMMB, (ii)URLLC, (iii) mMTC and (iv) hybrid	(i) MBRLLC, (ii)mURLLC, (iii) Human- Centric Services (HCS) and (iv) Multi-Purpose Services (MPS)
Information Timeliness	High	Extremely high

platooning, smart manufacturing (cloud-based robot control) feeder automation, health sector remote diagnosis), connecting drones for security, smart cities including video surveillance, wireless home entertainment (cloud-based gaming), social networking and personal AI assistant, etc. The Fig. 1 [3] illustrates the up gradations of various services from 5 to 6G.

The Global Mobile Suppliers Association (GSA) report [4] states that several countries have deployed or are in the process of the deployment of 5G and some are rolling out towards the advanced phases of 5G [5]. The Mobility Report of Ericsson [6] endorses the rapid growth of 5G. The report forecasts that by the year 2027, sixty per cent of the data traffic of the global mobile network is likely to be over in the 5G network. The various services that the telcos and communication service providers (CSPs) have launched, are the facilities of enhanced mobile broadband (eMBB), fixed wireless access (FWA), gaming applications and AR/VR-based services. In the enterprises, the various applications

of 5G are the smart factories, smart cities, ports and transport systems. While the rollout of 5G in many countries is still underway, researchers across the world are working to bring a new generation i.e. next generation wireless network (NGWN). A tentative timeline for the carrying out 5G, B5G, and 6G networks (standards) by international standardisation bodies is presented in the following Fig. 2 [7–9] considering the vision of 6G.

The potential benefits of AI/ML in the network will be noteworthy. It will enhance the performance of the radio interface, diminish the power consumption and significantly improve the end-user experience. Moreover, the integration of AI/ML will also play a vital role in 5G-advanced data collection and network management. To summarize, the deployment of AI/ML will help to find an improved performing network parameterization faster.

The main aim of the incorporation of ML/intelligence in the network is to support run-time intelligent decisions and estimate various network parameters. The deployment

Fig. 1 Upgradations of various services from 5G, to 5G Advanced and towards 6G

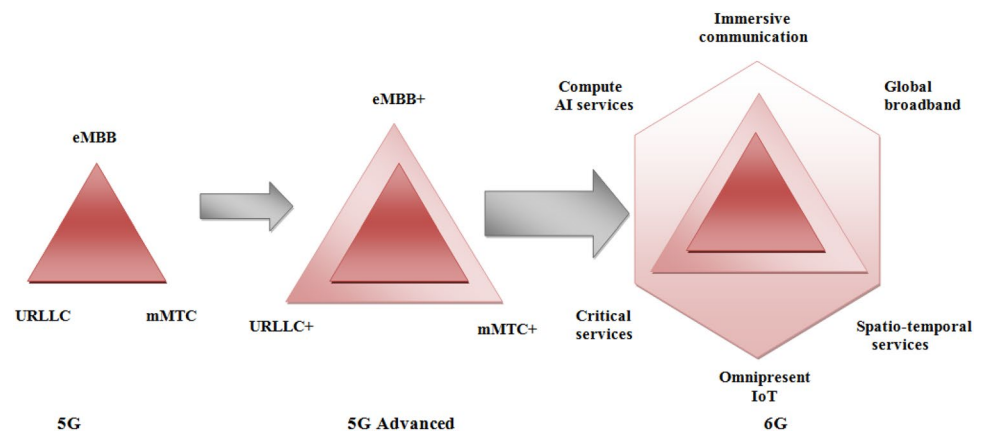
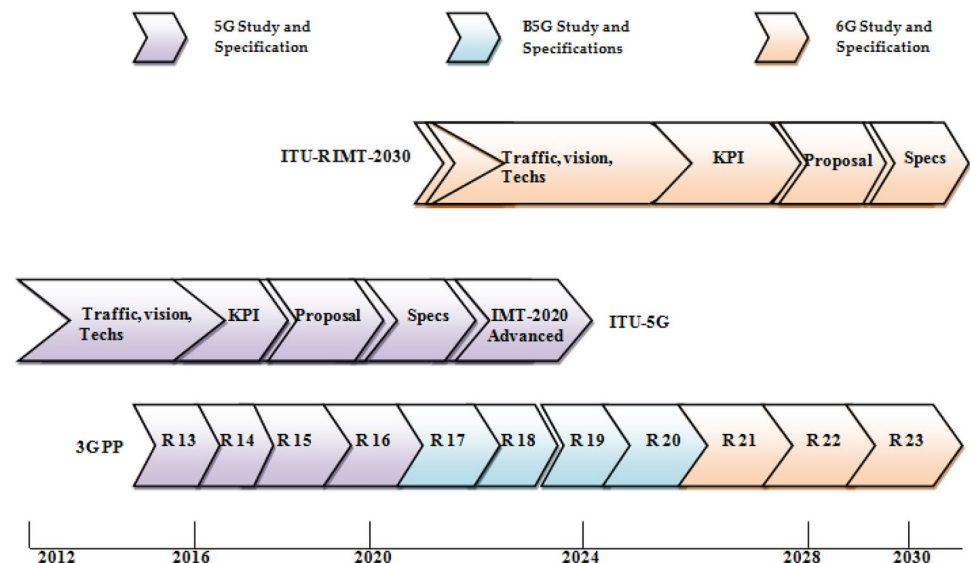


Fig. 2 Tentative timeline [7–9] for the carrying out of 5G, B5G, and 6G networks standards



of ML is strongly recommended as a potential solution for performance improvement (optimization) [11] in 5G and beyond (B5G). These include increasing network efficiency, critical network analysis and network slicing [12], etc. The aim is the implementation of the adaptive next generation wireless network (NGWN), and dynamic re-configurability in the network. The processes of dynamic reconfigurability at run-time are important fields of research in advanced 5G, beyond (B5G) and towards 6G [13, 14]. The integration of AI/ML in the telecom sector has a potential capability for replacement of conventional solutions (mathematical model-based/algorithmic) subject to the availability of computational power [15] and accessibility of adequate data. In 5G and 5G advanced, integration of Artificial Intelligence (AI) is suitable due to the benefits of its applicability and capabilities. The AI/ML has the capability of training the systems to carry out the tasks autonomously depending on the data-driven decisions. The important applications [10] of AI and ML are prediction, classification, categorization, recommendation, information translation and anomaly detection, etc.

The following Fig. 3 [16] and Table 1 [17, 18] portray comparative illustrations of target key attributes and Key Performance Indicator(s) (KPI) to be addressed for up gradations and rolling out from 5G advanced to 6G in the near future. Adopting network intelligence will significantly play a vital role in the telecom revolution towards network speed, mobility and various real-time applications.

Motivation

The various crucial factors [24] which drive the telecommunication industry to adopt intelligent networking/ AI are, (i) business growth addressing the growing demands of the

present and future market requirements; (ii) the need for dynamic scheduling and flexibility in resource allocation due to Infrastructure and Network Functions Virtualization (NFV), network slicing and associated technological advancements; and (iii) optimizing the operations and management [25]. Moreover, intelligent networking is considered an important tool [24] for potential solutions in the perspective of the telecom industries, as it can address urgent business needs and industrial issues. The AI/ML-based approaches play a significant role in network operations and management including intelligent radio access network (RAN) automation [26], intelligent data-driven solutions, different use cases of power optimization, etc. [27].

Moreover, the network data are treated as strategic gifts [28] for the telcos and the CSPs. Smarter utilization of the network data is the key to successful planning of the network, traffic forecasting, and efficient network management. This results in increased revenue and digital transformation. The approaches of AI/ML-based solutions (analytics) enable the CSPs to offer and perform automatic network operations and management and assurance (guarantee) of the services offered. Moreover, it offers guaranteed quality of service (QoS) and cost minimization. The impact of AI/ML adoption in the core network as observed by the service providers around the world has been reported in [31] and depicted in flowing Fig. 4. The figure illustrates that AI is playing a critical role in network performance improvement.

However, the incorporation of different AI/ML-based approaches cannot completely address the challenges associated with advanced 5G and B5G networks. This is due to the rapid growth of traffic, carrying heterogeneous traffic (classes) and time-constraint services. Addressing the aforementioned challenges, researchers are continuously concentrating on the developments of AI/ML integration in the advancement of 5G, Beyond 5G (B5G) [29, 30] and sixth generation (6G) networks and communication systems. Despite some progress related to the integration of AI/ML in the network, still there are many challenges to be addressed to incorporate network intelligence. Overcoming such challenges along with the inherited challenges of the B5G network is a process of rigorous monitoring and advanced analysis. CSPs will further focus on what has been indicated in Ericsson's recent report [31] and in [32]. This has been stated in the following Fig. 5.

The deployment of AI in NGWN is a critical issue for the Telco and CSPs. The following Fig. 6 illustrates the scenario of AI/ML integration challenges which have been reported in the 'Nokia AVA Telco AI and analytics' [28].

The aforementioned figures (Figs. 4 and 5) presenting the reports of the communication service providers (CSPs) indicate that there are strategic requirements to focus on different sectors of network intelligence by integrating AI/ML [13]. A lot of work is yet to be done for the

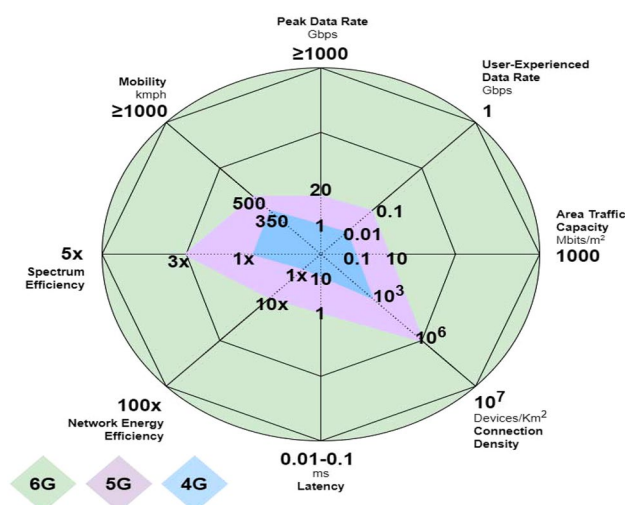
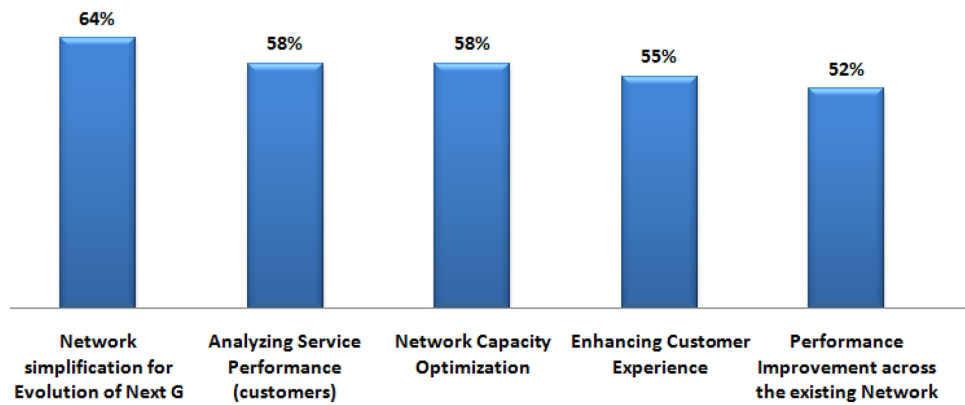
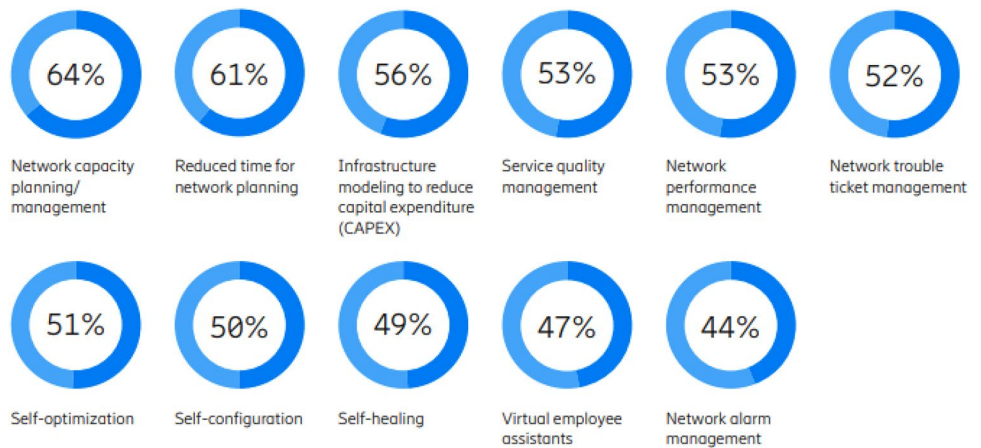
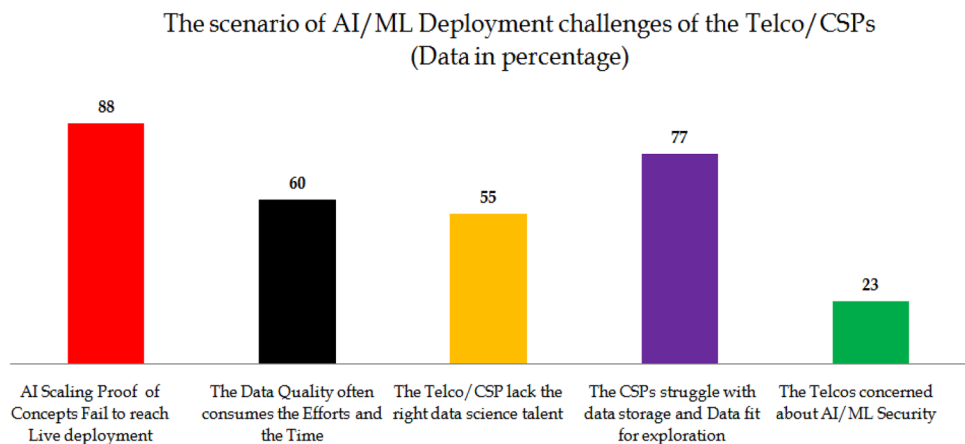


Fig. 3 Pictorial representation of comparative key requirements (performance) and KPI: Rolling out: 4G vs. 5G vs. 6G [19–23]

Fig. 4 Impact of AI/ML within the core network activities [31]**Fig. 5** The inclination of adopting network intelligence by the CSPs in various sectors to be focused further [31]**Fig. 6** Nokia AVA Telco AI and analytics: The AI/ML deployment challenges of the CSPs

proper deployment of network intelligence. Moreover, the incorporation of AI-based techniques boosts the return on investment (ROI) in NGWN.

Accordingly, investigations on recent state-of-the-art developments, future scope of work and trends for the deployment of intelligence in NGWN are significantly important [33]. The AI/ML-powered 6G network will play a significant role in the near future telecommunication

industry and associated sectors including the telecommunication-based manufacturing industries [34].

Key Contributions

In this section, the author discusses the significant and unique contribution of this work. Recently, a few survey-based papers [12, 21, 32–38, 40, 41] related to the

incorporation of ML-based approaches for network intelligence in NGWN have been reported in the literature. However, in these surveys, various important issues related to the developments, challenges and research trends of incorporating intelligence and various applications of ML in promising 6G network, edge intelligence, have not been specifically focused.

The unique and significant contributions of this paper are as follows:

- In this survey, a comprehensive study of the characteristics of the next generation wireless network, the background and the continuous developments, rolling out and deployments and the challenges involved from 5G to B5G and towards the 6G network have been discussed. The study has pointed out the recent progress of the telecommunication industries and future trends.
- An extensive discussion on the requirements of intelligence in the network has been discussed. A detailed survey of various state-of-the-art ML approaches in the contemporary telecom industry has been illustrated. Moreover, the paper explores various challenges of incorporating ML in the NGWN and the target applications.
- A comprehensive study on the various open issues and research trends focusing on extreme intelligence at the network edge have been illustrated. The paper illustrates the integration of federated learning (FL) in a promising 6G network. It investigates open research questions in edge intelligence (EI) in 6G.

The following Table 2 summarizes the contemporary surveys and compares them to the current work.

Structure of the Paper

The rest of the paper is organized as follows and is depicted in Fig. 7. Sect. "[The Challenges of next Generation Wireless Network](#)" presents a concise study of the challenges associated with the rolling out of the advanced /next generation 5G networks. In Sect. "[The Requirements of Network Intelligence: ML in Advanced 5G and Beyond](#)", the author illustrates the requirements of network intelligence: ML in advanced 5G and beyond. Sect. "[Typical Application Areas of ML in 5G/B5G : industry Perspective](#)" significantly highlights the application areas of ML in 5G and B5G in the context of cutting-edge technological developments in the telecom industry. The modern approaches of integrating various state-of-the-art ML techniques for target applications in advanced 5G and towards 6G have been discussed in Sect. "[State-of-the-art ML Integrating in Advanced 5G, B5G and Towards 6G](#)". The critical challenges involved in applying ML in NGWN have been investigated in Sect. "[Challenges Associated Applying ML in NGWN](#)". In Sect. "[ML](#)

[Trends in Future 5G and beyond](#)", the author highlights the research trends of the incorporation of ML in future 5G and beyond. Sect. "[ML Incorporation: Various open Issues Offer potential Research Directions](#)", explores various open issues and future trends related to the incorporation of ML in NGWN, which offers potential research directions. Furthermore, this section investigates the trends towards the extremely intelligent ML-powered 6G network, where the intelligence is to be incorporated at the network edge. Finally, the paper is concluded in Sect. "Conclusion".

The Challenges of Next Generation Wireless Network

This section introduces the various challenges of next generation wireless networks. With the rapid progress of 5G, the key research areas which have an impact on the advancement of 5G are to deal with high speed data and the stringent requirements of low latency traffic. These are the prerequisites of future 5G network and beyond. The research areas which have a significant impact are the deployment of dense cells, millimeter-wave communications, implementation of massive MIMO (MMIMO) [45], device-to-device (D2D) and machine-to-machine (M2M) communication. The various inherited challenges [46] associated with the design and development of protocols, multiple radio access technology (M-RAT), advanced coordinated multipoint (CoMP), coding methods, carrier aggregation, network virtualization implementation, integration with the cloud radio access networks (C-RAN), have an impact on rolling out towards advanced 5G and beyond (B5G) networks.

Furthermore, to address the challenges involved with the exponential growth of ever-increasing traffic, it is strongly recommended that the network capacity must be increased with a higher data rate. This could be possible by the implementation of the deployment of dense cells in a smaller area, the implementation of massive MIMO with beam forming [49] and the utilization of the millimetre-wave band. These challenges are discussed as follows.

Deployment of Denser 5G Cells

There is a critical challenge in the identification of cell sites and denser cell deployment in a small area to distribute the traffic and signal power enhancement. Small cells may be deployed within a small area, considering the issue of frequency reuse. Moreover, the deployment of such small cells will invite the challenges of frequent handoff issues or handoff failure. Fast-moving mobile devices may increase call drops significantly. Moreover, interference of the nearby cells and the unrestricted cell shape in a densely deployed 5G network are challenging concerns.

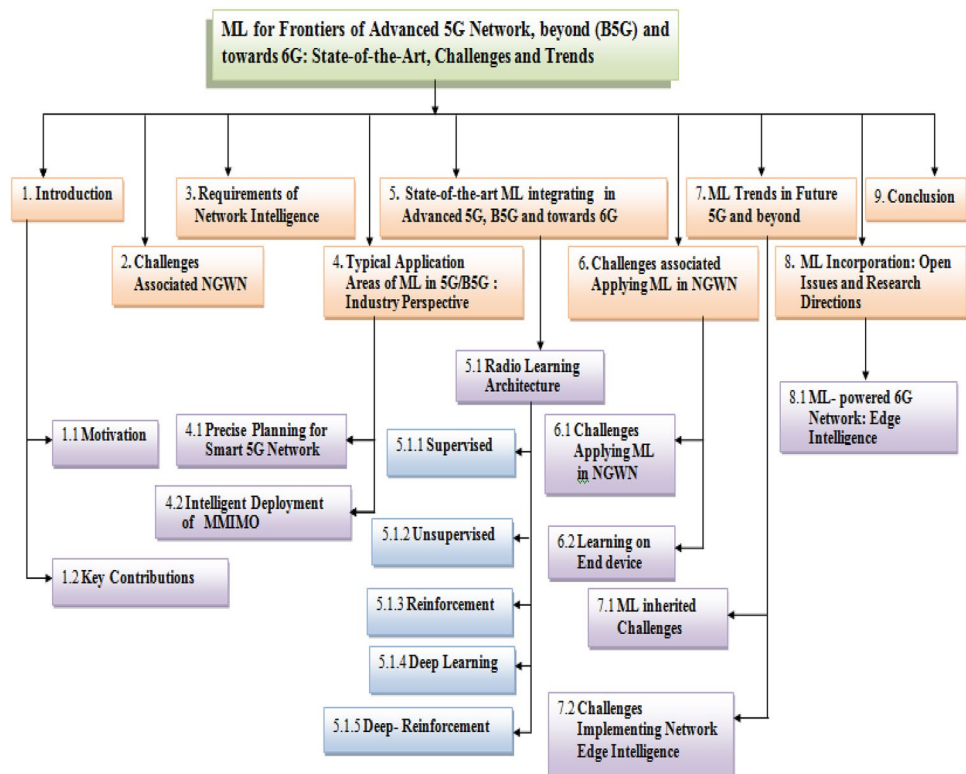
Table 2 Summary of contemporary review and surveys and comparison with this work

References	Publication Time	Domain / Major thrust areas and contribution	Performing key attributes of 5 G Advanced and 6G	Survey/ Reports of Telecommunica- tion Industry in the context of AI/ML: background, contem- porary developments and trends of the CSPs, Operators and Telcos	Deployment challenges of 5G Advanced in the con- text of ML integra- tion, Future research trends	Open Research issues related to integration of Edge Intelligence in 6G Network	Federated learning in 6G: Issues and Chal- lenges	Security issues related to incorporation of ML in NGWN
[12]	July 2022	Network Slicing for 5G Networks, applying ML in 5G, Mobile computing	×	×	✓	×	×	✓
[14]	Oct. 2021	6 G: Vision and Chal- lenges, sub networks	✓	×	✓	✓ (Limited focus)	×	×
[21]	April 2021	Frontiers of 6G: Requirements, trends, technologies, applications, future research	✓	×	✓	✓ (Limited focus)	✓ (Limited focus)	×
[32]	July 2022	Applying Reinforce- ment Learning (RL) in 5G and B5G, network slicing and virtualization	✓	✓ (Limited focus)	✓	×	×	✓
[35]	April 2017	ML for NGWN, Review of various ML techniques	×	×	✓	×	×	×
[36]	April 2022	Rolling out from 5 to 6G, challenges, Technologies and Applications	✓ (Limited focus)	×	✓	✓	×	✓ (Limited focus)
[37]	Dec. 2020	Vision and require- ments of 6G, ML incorporation	✓	×	✓	✓	×	✓
[38]	Feb. 2021	Deep Learning (DL) in 5G, MMIMO, channel coding, multiple access, resource allocation and network security	✓ (Limited focus)	×	✓	×	×	✓

Table 2 (continued)

References	Publication Time	Domain / Major thrust areas and contribution	Performing key attributes of 5G Advanced and 6G	Survey/ Reports of Telecommunication Industry in the context of AI/ML: background, contemporary developments and trends of the CSPs, Operators and Telcos	Deployment challenges of 5G text of ML integration, Future research trends	Open Research issues related to integration of Edge Intelligence in 6G Network	Federated learning in 6G: Issues and Challenges	Security issues related to incorporation of ML in NGWN
[39]	June 2022	ML in Next Generation Wireless Network	✓ (Limited focus)	×	✓ (Limited focus)	×	×	✓
[40]	Aug. 2022	DL approaches towards 6G, cell-free MMIMO	×	×	✓	×	✓	×
[41]	Aug. 2022	IoT with the integration of AI/ML	✓	×	✓ (Limited focus)	×	✓ (Limited focus)	✓
[42]	April 2023	ML in 6G Ultra Reliable and Low Latency Services	✓	✓	×	✓	×	✓ (Limited focus)
[43]	September 2023	Edge Learning in 6G enabled IoT	✓	×	×	✓	✓ (Limited focus)	✓
[44]	October, 2023	FL-based AI models for B5G/ 6G for QoE forecasting in a vehicular network	✓ (Limited focus)	×	×	✓ (Limited focus)	✓	×
This Work		ML techniques for Advanced 5G Network, beyond (B5G) and towards 6G: State-of-the-Art, Challenges and Trends	✓	✓	✓	✓	✓	✓ (Limited focus)

Fig. 7 Organization of the paper: main components of this survey



MMIMO Along with Beam Forming

Massive Multiple Input Multiple Output (MMIMO) [45] and beam forming play an important role in advanced 5G network. The challenge is due to the pilot contamination, with a large number of antennae from the adjacent pico cells. The challenge is to optimize the pilot orthogonality [46], in such a technique that the resources should be least consumed. Moreover, there is a challenge in the run-time process of channel estimation with the growing number of antennas (TDD). An advanced algorithm is required for proper channel estimation (FDD) and to minimize the overhead (signalling).

Challenges Associated with Minimizing of the Network or System Latency

5G and B5G need a stringent requirement of latency minimization with the advancement of cloud-based applications. These include tactile Internet and real-time gaming, etc. The challenge is provisioning the QoS parameter: latency lower than 1 ms [17, 20]. However, this will be a concern for the design and development of the other network layers.

Improvement of Spectral Efficiency

To guarantee the data delivery at an ultra-fast speed, the Spectral efficiency should be increased [47]. Orthogonal

Frequency Division Multiplexing (OFDM) plays an important role in combating the issues of Inter Symbol Interference (ISI) [48]. However, OFDM invites several concerns associated with a high peak-to-average-power ratio (PAPR). This downgrades the efficiency of the installed power amplifier in the system.

The Requirements of Network Intelligence: ML in Advanced 5G and Beyond

Incorporating artificial intelligence in the network is aimed at providing for the need for automatic network management, self-optimization or self-organizing Network (SON) [50] and network adaptability. Deployment of ML supports network auto-configuration and overall network performance improvement. The ITU-T steering committee [51] advocates for the solution of ML integrations in advanced 5G and beyond. The objective of integrating intelligence is to offer the connection requirements for a large number of devices/user equipment(s) (UE), higher throughput, and enterprise applications. Moreover, there are several concerns related to network monitoring and management due to the fast growth of network traffic.

The advantage of deployment of network intelligence by incorporating ML is to address complicated network management and network optimization problems (NP-hard). The advantage of the use of ML is that it extracts systematically

the required information from the incoming traffic data. Moreover, ML discovers the patterns automatically for appropriate decisions. As a result, service providers/network operators will be able to make suitable run-time decisions with ML tool-assisted analytics [26] along with knowledge discovery. As pointed out by the authors in [52, 53], the initial studies regarding 6G, highlight that the various applications of ML will play a key role in NGWN.

Moreover, applying machine learning (ML), network load balancing and load forecasting, and traffic analysis of slice-based networks [54] can be executed at run-time. Machine learning (ML) is the collective technique that is used to learn, automate and optimize the discriminative features using a large volume of data set in a system or computing platform. Here, the human intervention for processing is unrealistic. Without the incorporation of AI and ML techniques, it is next to impossible for the service providers/telcos to cope with the various critical challenges of rapidly developing applications of 5G and beyond. In such applications, huge volumes of data of the network are required for processing and analysis.

Furthermore, intelligence in 5G is required [55] to address the challenges involved due to the complexity of the network. The influencing factors of network complexity are as follows. The first is the deployment of various types of end user devices and intermediate internetworking devices. The second is the coexistence, integration and upgradation of internetworking components associated with continuous technological transformation or rolling out from 4 to 5G and towards 6G. At present, many countries are struggling to work out such rolling out for up gradation of their existing network infrastructure of 4G to 5G [56]. The third is the advancement of cloud computing, virtualization, slicing of the network [57] and mobile edge computing (MEC) increases the complexity of the network in a greater way. Moreover, traffic management, continuous network monitoring and analytics [26], and enforcement of the traffic policies by the CSPs increase the network complexity.

In addition, processing various traffic classes and data of voice, Ultra Reliable Low Latency Communication (URLLC) and video surveillance, etc. increases the network complexity. Furthermore, real-time traffic monitoring, testing and troubleshooting make the network congested, resulting in increased complexity in the network. To optimize the network performance it is important to dynamically estimate the parameters and to make intelligent decisions at run-time. The aim is to address various 5G deployment challenges and to provide target quality of service (QoS), customer satisfaction (QoE), resource utilization, efficient network management, and administration.

To address various challenges of complexity and heterogeneity of network infrastructure, machine learning techniques work as an efficient tool for service providers in a

congested and dense network. In such a network, there is a requirement to process volumes of data. The various state-of-the-art approaches of ML can learn the parameters at run-time from the network system environment. ML models can predict the future system environment and make proper decisions to adopt the rapidly changing network environment.

AI and ML help CSPs deliver services in a smarter way. To handle the 5G network system and its components there are categorical requirements for the computations of multiple parameters in a highly complex and multi-varying system environment. The need to learn various patterns requires different categories of problems within each component of the learning model.

The integration and applications of various ML algorithms in 5G and B5G networks.

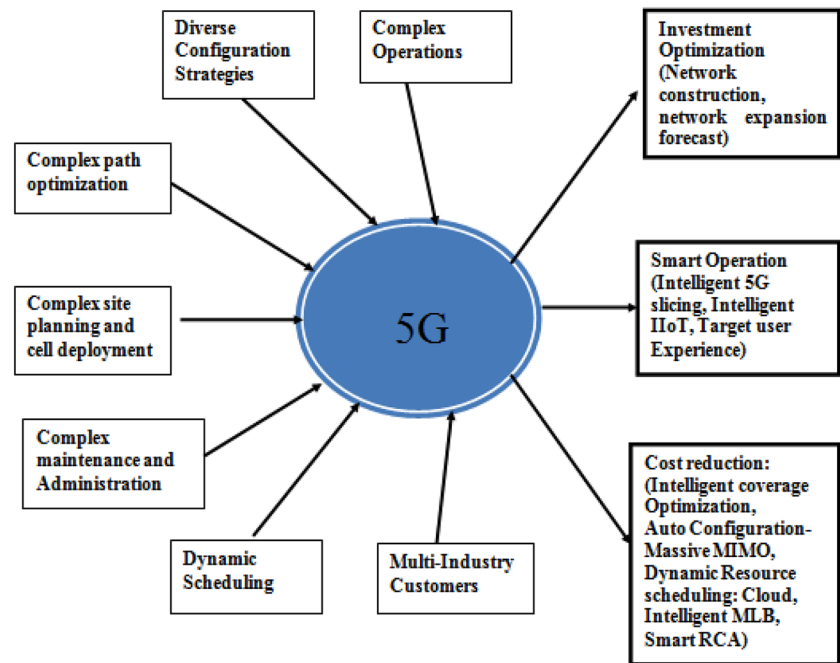
(systems) are largely categorized [35] into (i) Supervised Learning (SL), (ii) Unsupervised Learning (UL), and (iii) Reinforcement Learning (RL). Moreover, the focus is on other classes of ML algorithms, such as Heuristics-based and Markovian models. These are being applied for different 5G and Advanced 5G network and applications.

Typical Application Areas of ML in 5G/B5G: Industry Perspective

This section introduces various application areas of machine learning in 5G and BG networks. The ML in the network can be applied for parameter estimation and performance improvement. It can be applied to financial metrics, such as cost per bit, resource utilization, reliability, latency, jitter, throughput, and Quality of experience (QoE) improvement. The deployment of ML can be done at the network edge at run-time using lightweight AI/ML engines [58] for real-time (time-bound) computation. This real-time computation is strongly recommended for IoT-integrated 5G and B5G [40, 41] networks, where low-latency applications (processes) run. Centralized deployment of ML in the network system is generally used for high computation. ML at the edge devices in the network are used for quick run-time decisions. Moreover, ML deployment at the edge has the advantage of lower computational complexity. The following Fig. 8 illustrates the representative challenging application scenarios of network intelligence utilizing ML in 5G considering the industry perspective.

Furthermore, the incorporation of AI/ ML in advanced 5G can enhance system performance, network planning, and forecasting network capacity. Moreover, it supports smart multiple input multiple output (MIMO) implementation, dynamically auto-optimizing the network coverage area, dynamic resource scheduling, smart slicing, integration with cloud network (dynamic), etc. In addition, AI/ML enhances Operation and Maintenance (O&M) efficiency resulting in

Fig. 8 Typical Application scenarios of Intelligence (AI/ ML) in 5G: Industry Perspective



cost reduction and overall investment optimization of the communication service providers (CSPs).

Precision Planning for Smart 5G network

The telcos face major challenges in the deployment of cell sites in a 5G network. These challenges arise due to complex scenarios of handling multiple systems and managing different classes of traffic including mission-critical and multi-frequency bands. In such cases, the ML can be an effective solution as it can analyze the multi-dimensional data and can predict the peak of the incoming traffic, and resource utilization in a cellular region. Furthermore, the interference and multi-mode coverage can be measured using ML/AI with an aim for the estimation and reconfiguring of network parameters and performance optimization. These reconfigured and optimized values can be applied as recommended values for further network planning, to increase the capacity of a pico-cell and overall network system capacity, etc. This may outcome convenient network planning and installation of cell towers along with optimal deployment with a reduced budget.

Intelligent deployment of Massive MIMO

Multiple Input Multiple Output (MIMO) technology uses multiple antennas to transmit and receive signals in the wireless communications. The massive MIMO (MIMO) [59] is the key requirement of 5G deployment [60]. For optimal user experience and coverage, the beam forming of MIMO channels [60] needs to match the cell users' distributions.

Furthermore, the interferences from the channels (broadcast) of the other cells (neighbouring) must be minimized. The quality of coverage and the efficiency of the network is improved by the beam forming which is affected by the weight setting and tuning (adjustment) using ML. In 5G, the challenges are the requirements of addressing different covering areas. The telcos can integrate ML to monitor the continuous changes in user distribution. Moreover, the deployment of ML in the network will dynamically adjust (fine-tune) the weights and predict the distribution by learning from the observed patterns. This could be the possible solution.

State-of-the-art ML Integrating in Advanced 5G, B5G and Towards 6G

The Machine Learning (ML) is a type of subset of Artificial Intelligence (AI), in which machines can able to learn, can perform and optimize their operations by utilizing the knowledge (operational) and the experience acquired in the form of the data [61]. More investment in machine learning integrated network i.e. incorporation of network intelligence implies better network performance [17]. The Machine learning models are applied to learn the patterns or characterises. These models are applied in the tasks, for example, regression, classification and interactions of an agent in an environment. Depending on the type of available data and the explicitness of the objectives of learning, in general, ML can be categorized into three major variants. They are, categorically: (i) supervised, (ii) unsupervised, and (iii)

reinforcement learning. Recently, diverse approaches of ML have been applied for continuous performance upgradations [35] of the next generation wireless network (NGWN). The following Fig. 9 depicts the inter-relationship among Artificial Intelligence (AI), Machine Learning (ML), deep learning (DL), Supervised Learning (SL), Unsupervised Learning (USL), Reinforcement Learning (RL) and Deep Reinforcement Learning (DRL) in the perspectives of NGWN and applications.

Radio Learning Architecture

The various methods of machine learning are being integrated for modelling and addressing the technical challenges in advanced 5G network, B5G and 6G. The target implementation of intelligence is in MIMO (in large-scale), device-to-device (D2D) networks and heterogeneous networks

consisting of a large number of femto cells. The following Fig. 10 illustrates ML radio learning techniques and their target applications in the network.

In the subsequent section, the author presents the various classes of ML algorithms and their variants, state-of-the-art approaches and potential target applications. Figure 11 illustrates the simplified working flow of Supervised Learning (SL), Unsupervised Learning (USL), and Reinforcement Learning (RL).

Supervised Learning (SL)

In supervised learning (SL) algorithms, the training is performed by applying labelled data sets [35]. In these approaches, the system knows the input and the desired output data to be predicted. In these approaches, adequate data are required to apply effectively in target applications.

Fig. 9 Relationship of AI, ML, DL, SL, USL, RL and DRL

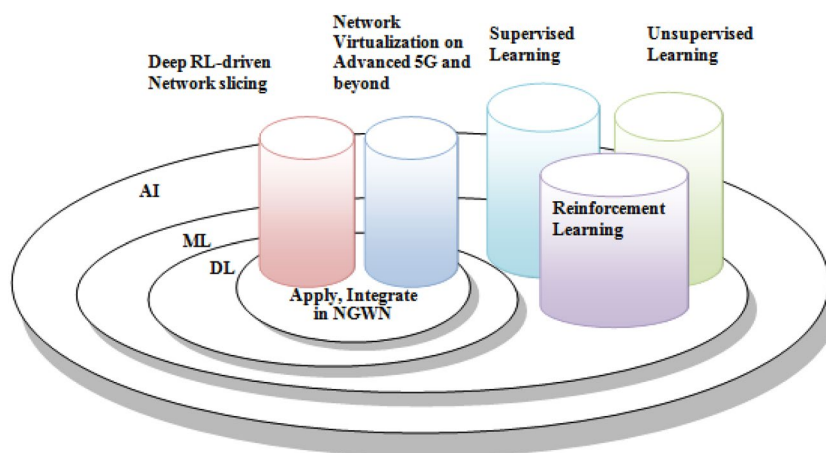


Fig. 10 Radio learning Architecture: Incorporation of ML/Intelligence in network

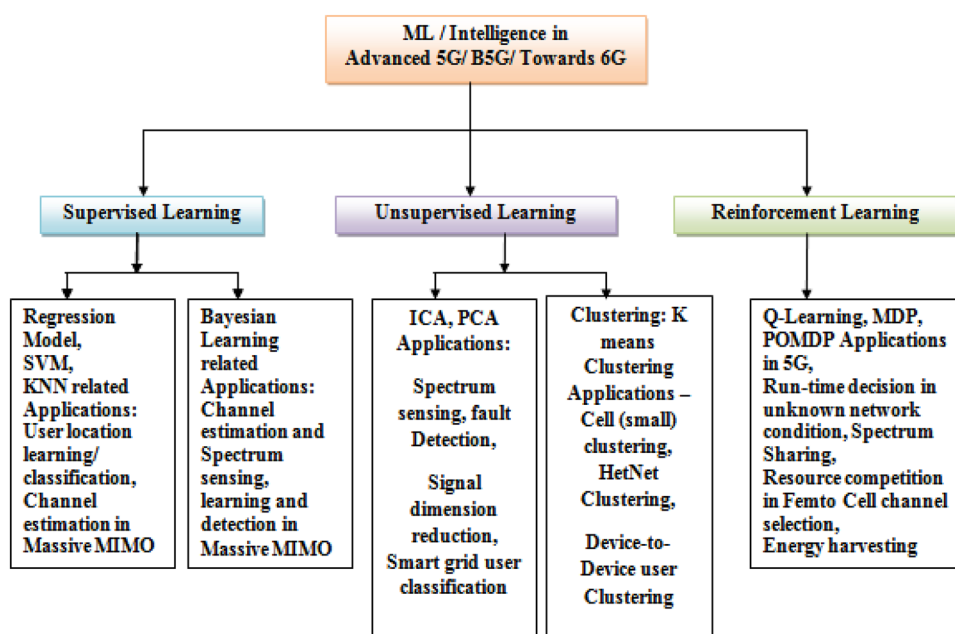
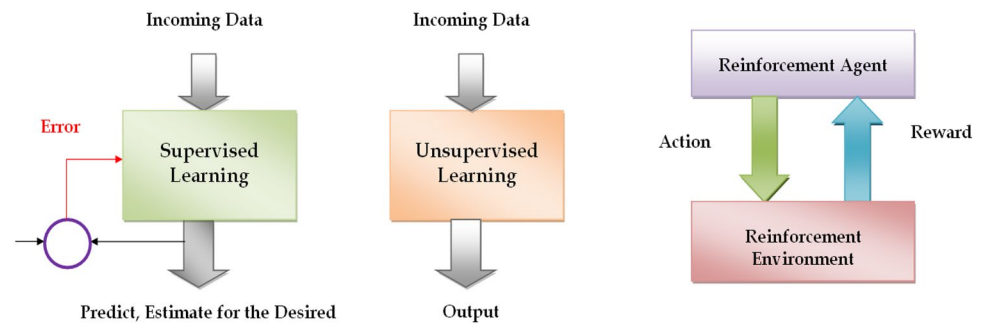


Fig. 11 Supervised, Unsupervised and Reinforcement Learning

The supervised learning methods are dependent on labels and known models. The SL algorithms can be applied for network parameter estimation. The typical applications are channel estimation in MIMO, detection of data, sensing the spectrum [62], adaptive filtering, and detection of the white space (cognitive radio) in 5G communications. Moreover, these learning techniques can be applied in applications related to the higher layers. These are finding out the mobile users' locations and behaviour. This helps the operators for the improvement of overall network and system QoS.

In the following Table 3, the author summarizes the state-of-the-art supervised ML approaches for potential target applications in B5G and 6G.

Unsupervised Learning (USL)

The applications of unsupervised learning (USL) algorithms in the telecommunication network depend on the input data pattern (incoming traffic pattern) itself heuristically. The application areas include the clustering of base stations in the heterogeneous networks (HetNets). These are load balancing and clustering of cells in small-cell networks

Table 3 The various methods of Supervised ML and their applications in B5G and 6G

References	ML Approaches/ Methods	Target Applications
[63]	Artificial Neural network (ANN)	Bandwidth allocation dynamically, minimization of latency and overall performance improvement
[64]	K-Nearest Neighbour (KNN) and Random Forest	In a mm wave system: path loss prediction and optimization
[65]	ANN and Convolutional Neural Network (CNN)	Channel estimation and prediction of CSI in MMIMO
[66]	Mapping Mode applying Supervised learning (Deep)	Channel mapping in MMIMO: mapping in space, frequency domain. It minimizes the overhead due to training and feedback
[67]	Deep Neural Network (DNN)	Users' requirements prediction in dynamic networks
[68]	knn with Support Vector Classifier (SVC)	Address issues (multi-class problem) related to beam-selection in communication (mm-wave) system
[69]	Recurrent Neural Network (RNN) and DNN	Prediction of load (traffic) and power allocation –optimization
[70]	Gated Recurrent Unit (GRU)	Beam prediction optimally and coverage improvement in drone-based applications
[71]	CNN	For classification requirements and identification of malware traffic
[72]	DNN	Transmitter power prediction and intelligent beam forming
[73]	Support Vector Machine (SVM), ANN	Transmit power prediction along with the boundary decision intelligent classification
[74]	SVM	Applied for Beam selection sum-rate, Attains a higher value of the Average Sum Rate (ASR). The computational complexity must have a significantly lower value
[75, 76]	ANN, SVM, Random Forest	Data rate prediction with a target of minimum error (prediction)
[77]	DNN	Applications related to caching: for optimization of caching in IoT
[78]	SVM	In UAV systems: Applied for the detection of jamming and spoofing. Identification and detection of intrusion attacks
[79]	ANN	In UAV system: For the detection of GPS signal (spoofing) and for the minimization of false alarm
[80, 81]	Ensemble with ANN	For RSS prediction (accurately) in UAV systems

(ultra-dense) in a cooperative manner. Moreover, the applications are end users' behaviour categorization, intrusion detection, detection of network faults, etc. Unsupervised learning is applied for the extraction of the features and their classifications, the estimation of the distribution, and generating distribution-specific samples.

The following Table 4 summarizes the state-of-the-art USL models and their variants for potential target applications in B5G and 6G.

Reinforcement Learning (RL)

Reinforcement learning (RL) is executed depending on an indicator (feedback performance) or the reward learnt from the environment. This learning is performed after computing a precise output for a definite observation by converging (adaptively) to the desired best performance through the increment of the reward. RL is a method of negotiation form

of supervised and unsupervised learning. In RL, the earlier understanding of the best performance of the system gives indirect supervision. There is no offered training data associated with the target output. Reinforcement learning is a dynamic learning (iterative) and decision making process which can be applied in advanced 5G networks and beyond [15]. These types of learning approaches are useful to solve the problems of sequential decision making. Sequential decisions making are reliant on Markov decision processes [100]. Moreover, the learning techniques can be used for appropriate decisions of mobile users in an unknown network environment.

Classically, RL is applied for the solutions of classification and control problems. The various potential applications are power control with optimization (network physical layer); spectrum sharing; and the channel access procedure in an unknown channel availability. Moreover, RL can be applied in densely deployed femto cells [101],

Table 4 Various approaches of Unsupervised ML for target applications in B5G and 6G

References	ML Algorithms/Approaches	Target Applications
[82]	Hidden Markov Model (HMM)	Applied for the precise prediction of the user location considering the mobile network as a graph (state-transition based approach)
[83]	Discrete Markov chain Model (DMCM)	Estimation of user mobility, prediction of the forthcoming cell where a user is likely to move into i.e. movement prediction based approach
[84]	Association algorithms related to the Unsupervised UE	Applied for the optimization of the data rate, traffic load in RF, THz characterized telecommunication system and network
[85, 86]	CNN and Long Short-Term Memory (LSTM)	In modulation recognition to achieve target performance improvement
[87]	Self-organizing Map (SOM)	Applied for the fault identification and recovery from the fault
[88]	(i) k-means clustering and Fuzzy C-means clustering, (ii) Kohonen's SOM	Applied for efficient fault detection
[89]	DetNet	Efficient channel Estimation with lower time (computation) spent
[90]	PC algorithms	In a D2D communication: optimization of power control, deals with the computational complexity, resource allocation and system throughput improvement
[91]	Gaussian Mixture Model (GMM)—Unsupervised	For the enhancement of physical layer security
[92]	Q-learning and k-means clustering	Deals with spatially clusters (in correlated) and users, places the UAV in (3-D way) an efficient way
[93]	Bayesian approach (non-parametric)	Addresses security issues: malware detection and the access control applicable in IoT-based systems
[94]	Unsupervised CNN and Sparse Auto Encoder (SAE)	Intrusion detection with greater degree of accuracy
[95]	K-means	Applicable in Non-Orthogonal Multiple Access (NOMA) enabled system: addresses the issues regarding user selection, focuses optimization of power in the system
[96]	K-means	Addresses the issues of user selection, accurate optimization of power in NOMA –integrated systems
[97]	k-aware k-means (combined)	For fault Detection: self optimizes the k-value, further it precisely identifies the inconsistencies in the system
[98]	Deep Reinforcement Learning (DRL): Unsupervised trained	Identifies the chances of attacks in 6G enabled systems
[99]	Deep Neural Network (DNN)	The sum-rate maximization in MIMO. This is applicable for a single base station (system) subject to the condition that the computational speed has been improved significantly

and 5G Radio Access Network (RAN) slicing [102]. RL is applied for the allocation of the resource in a distributed manner and energy harvesting networks, etc.

In the following Table 5, the summary of various state-of-the-art Reinforcement Learning (RL) models and their variants are presented for target applications in B5G and 6G.

Deep Learning (DL)

Deep learning (DL), a type of neural network (having 3 or more layers) is a subset of machine learning [15]. These neural networks try to simulate the actions of a brain (human) and apply learning from large amounts of data [125]. Recently, it has been observed that there is potential growth of applications related to the Deep learning, methods based

Table 5 Approaches of incorporation of RL and its variants and methods for target applications in B5G and 6G

References	ML Algorithms/Approaches/ Methods	Target Applications
[103]	Q-learning and Deep Q-learning	Energy saving applications in cooperative networks
[104]	Deep Reinforcement Learning (DRL) -based approach	Performs intelligent management of the resource by applying run-time intelligent decisions on UE-based communication systems. Power consumption is reduced
[105]	Neighbor-Agent Actor Critic (NAAC)	A distributed spectrum allocation framework based on multi-agent DRL: Utilizes the information from the neighbours of the user and performs improvement in 'sum rate' in a D2D communication
[106]	Actor-critic	Intelligent resource allocation, joint user control, IoT based networks
[107]	Q-learning	In resource allocation: It reduces the outage probability by allocating the resources (of channel) subject to the condition that the power constraint (average) must be satisfied at the node (energy harvesting mode)
[108]	Reinforcement Learning (RL) model (Offline-based)	In Handover: Optimization of the handover decisions
[109]	Q-learning (Deep)	Applications of lower energy consumption and lower value of convergence time in cooperative communication
[110]	Single-agent Q-learning (SAQ-learning) and Bayesian learning automata (BLA) based multi-agent Q-learning (MAQ-learning)	Applied for efficient resource allocation and offloading (with higher degree of accuracy) in network
[111]	RL-CAT and RL-pCAT	Increasing data rate for uplink along with downlink
[112]	Q-learning	Selection of channel with higher degree of accuracy and the probability of detection, minimizes the access delay along with overhead occurs due to scanning
[113]	Q-learning (Deep)	CSI and QoS-based adaptive power allocation (dynamically)
[114]	MAMAB	Applications related to caching, online learning of caching techniques
[115]	Generative Adversarial Network (GAN) Double Deep Q Network (DDQN)	Allocation of spectrum for each network slice and Q-learning-based technique for the optimization of spectrum allocation
[116]	Multi-agent RL, DNN	Power control operations and throughput maximization in IoT-based network (energy harvesting mode)
[117]	Deep actor-critic	Applications related to caching, performs for the optimization of cache hit ratio and minimization of transmission delay
[118]	Q-learning (multi-agent based approach)	Optimal power allocation
[119]	Double DQN and Distributed resource allocation model	Applied for resource allocation and minimizing the latency
[120]	DRL	Addresses security issues: handles attacks due to jamming, performs for throughput improvement
[121]	RL (Distributed)	Addresses the frequent handover in mm wave-based systems and the signal overhead is minimized
[122]	Deep Q Network (DQN): multi-agent and multi-user based	Applied for optimal power distribution
[123]	Novel RL	Resource allocation subject to the condition of the latency constraint in a V2V communication
[124]	Deep Q-learning	Applied for significant fault detection

on artificial neural networks (ANNs) to solve many learning problems in telecom sectors [125]. Deep Learning (DL) considers a rigorous system structure for the required representing and learning structures (correlation) in the available data by executing in a supervised, unsupervised, reinforcement, or applying in a hybrid fashion/mode. The various approaches include Multilayer perceptrons (MLPs), Convolutional neural networks (CNN), Recurrent Neural Networks (RNN), etc. However, the challenges are due to the issues associated with dimensionality [126]. These types of learning require a large volume of datasets (training) to achieve considerable performance improvement. Furthermore, the heterogeneity of the network is a challenging concern in applying DL due to the coexistence of various network operators [13].

Deep Reinforcement Learning (DRL)

Deep Reinforcement Learning, a blend of RL and Deep neural networks (DNN) is an excellent performer [127] in telecommunication networks. DRL is an alternating

solution to statistical models applied to address the network slicing challenges. In this type of learning, the interactions are executed in a trial-and-error process [128]. Learning from the actions is performed to improve the rewards. In DRL, there are certain drawbacks of poor performance associated with the dimensionality in a bulky state-action space. Applying RL, in combination with the Deep Neural Network (DNN) is the potential solution to overcome the challenges.

In addition, intelligent algorithms, such as swarm intelligence, methods of neural networks, neuro-fuzzy methods, and evolutionary algorithms can be applied for network performance improvement. However, the methods must consume the least time for appropriate decision (parameter estimation) at the run-time in a diversified network environment. The approaches based on neural network and deep learning are applied for performance improvement. These include handoff decisions [129], prediction [130], and spectral white state estimation [131].

The following Table 6 lists the summary of recent DL approaches and their variants for potential target applications in 6G.

Table 6 Approaches of Incorporation of DL/DRL and its various methods/ variants in 6G

References	ML Algorithms/ Methods	Target Applications
[132]	Deep Reinforcement Learning (DRL)	A DRL based framework for efficient hand over mechanisms in a 6G enabled optical wireless communication systems. DRL applied the Q-target as well as Q-evaluation for training and updating the neural network
[133]	Deep Reinforcement Learning based Recruitment (DRLR)	A DRL based DRL recruitment (DRLR) method applicable in unmanned aerial vehicles (UAVs) for the data collection from the deployed sensors in a 6G enabled IoT network
[134]	Feature-aided Adaptive Tuning-Deep Learning (FAT-DL)	Massive detection of the devices in 6G applying FAT-DL an approach of layer by layer design (training). The trained data is used for the decision of the parameters (distribution) of the network devices
[106]	Hierarchical Deep Actor-Critic-Reinforcement Learning (H-DAC-RL)	An approach of 6G space with terrestrial system incorporated with IoT. The hierarchical deep actor–critic RL proposed for resource allocation along with the operations of the network control
[135]	Adaptive Observation Window for Deep Q Network (AOW-DQN)	An ML based architecture for the 6G enabled IIOT for the engancement of learning efficiency for a deep-Q network
[136]	IScaler	A DRL based method for scaling of the resource along with the service placement in a mobile edge computing platform in a 6G based Internet of Everything (IoE) systems. This method is referred as: IScaler. The system offers improved scaling, placement decisions in a challenging network environment
[137]	Deep Deterministic Policy Gradient (DDPG)	The network slicing (6G RAN) technique is executed applying DRL—based Markov decision. The proposed method is dependent on the deep-Q-network along with the method of deep deterministic policy gradient
[138]	Micro-Safe	A Deep neural Network- based intelligent 6G dependent transportation systems to minimize the accidents and to maintain safety services for the end-users with accurate decisions

Challenges associated applying ML in NGWN

In this section, the author discusses various challenges of machine learning in NGWN. Despite advantages, there are certain pitfalls and challenges associated with the applicability of ML in 5G and beyond 5G network (B5G). These are illustrated below.

1. Deployment of appropriate learning processes for network performance improvement is a challenging task compared to the existing state-of-the-art deterministic models. Learning is a continuous process (incremental type) of non-linear progress. There is a high possibility of striking down the local minima. The network performance may be affected locally, and then the network resumes its performance improvement.

2. The prediction may be inaccurate for the black-box perception of AI/ML while processing large data volumes.

3. Model predictions/ learning may have a higher variation of results in comparison to deterministic network models. There is a challenge to apply appropriate ML techniques. In general, there is a trend that the network operators preferably apply a restricted approach of supervised learning (model) followed by other ML techniques to avoid any undesirable deviation from the expected results. This approach ensures a greater degree of accuracy of the desired outcome.

Challenges of Implementing Learning Process in 5G

With the fast development of 5G standards, pre-commercial tests [139], and the rolling out [140] of the advanced 5G networks in several countries, telcos and service providers are facing diverse challenges. These challenges are network complexity, diverse services, handling heterogeneous traffic classes, and end-user QoE requirements.

- **Network complexity:** The complexity of densely distributed 5G networks arises from complex site planning and installation.
- **Antenna Array:** Densely deployed antenna arrays (large-scale) make the system complex.
- **Scheduling:** Complex scheduling algorithms are necessary for resource utilization in the network.

With the rapid transformation from centralized to distributed network architecture, the network operator faces challenges in applying appropriate places in the 5G network with proper learning techniques to ensure network performance improvement. Initially, a centralized AI/ML system was applied in the cloud platform. As the learning is done offline in a supervised mode, the location has

no impact and it solely depends on the availability of resources and cost. On the contrary, with the application of automation (real-time) in unsupervised closed-loop learning, it is very important and challenging to find the appropriate location in the network. This is because the network operators must focus on real-time low latency, guaranteed bit rate (GBR) traffic [141] and location-dependent tasks. The critical decision of location selection is dependent on various deciding factors and is dependent on a case-to-case basis. Without compromising operational constraints, availability, cost, and network security, in general, an edge location is a preferred choice for SON [50]. Furthermore, edge processing is preferred for applications/ traffic of delay (latency) and jitter-sensitive traffic.

The deployment of ML for intelligent learning at run-time in a dynamic network environment considering the network reliability, scalability and security aspects, subject to the constraint that the deployed or the applied ML techniques must ensure to minimum network computational overhead, processing delay and thus resulting in minimum congestion. Therefore, working on optimal edge location with the incorporation of ML is drawing attention and is a challenging research area.

Learning on the *end device* in the 5G Network

To offer an excellent QoE, it is suggested that the initial learning rate can be performed centrally (independent of a specific end user device). Eventually, the learning can be done at the end user device. Accordingly, there arises the challenge of inter-dependency and co-existence of edge ML, cloud ML and device ML altogether. The aim is that intelligence is distributed all over a network without even any centralized control.

ML Trends in Future 5G and Beyond

Incorporating with AI/ML, future 5G and beyond, offers several research directions. Several works are being focused related to the implementation of network intelligence with an aim to minimizing the network latency and processing delay, processing of deadline-driven traffic, monitoring the network environment and updating parameters at run-time, for the requirements of network adaptability, smart load balancing, intelligent routing and handoff decisions, processing of data in real-time in the network, traffic pattern analysis and forecasting, smart processing of heterogeneous traffic, auto-recovery in case network failure and overall smart network decision at run-time. Both of the challenges involved with ML and NGWN will have to be focused on prime importance and offer future research directions.

ML inherited Challenges

There are several inherited challenges associated with ML. The challenges of the applicability of ML for predictive analysis [142] are relevant in next generation wireless network (NGWN). These are discussed in the following section.

Vulnerability of Errors

ML is highly vulnerable to both human and machine errors. If a sufficient amount of data is not trained, then there is a high probability of leading to a biased prediction. The quality and the accuracy of prediction are mostly dependent on the correctness of the dataset (training). In ML, the prediction errors are difficult to identify and rectify due to algorithmic complexities [143].

Issues with Acquisition of Data

The learning and prediction in ML are dependent on data, associated patterns and the acquisition of data. This is not a simple process. The larger amount of data offers better reliability in the prediction process [143].

Time Requirement

ML requires adequate time for training and learning for proper functioning. Applying ML for predictive analysis requires time and patience for accuracy. A huge volume of data for processing may increase the time complexity and a greater amount of processing time requirement.

Challenges Involved for Verification

It is difficult for required verification of specific facts, which may not be included in the historical dataset. This may result in erroneous predictions applying ML in some cases. Furthermore, the model may consume a significant amount of time for training and mastering the dataset.

Challenges of Implementing Network Edge Intelligence

With the rapid up gradation and a large number of devices connected in 5G, there is a major need for distributed intelligence in the network. The telcos are focusing on addressing various research challenges associated with the rapid transformation from centrally cloud-based learning to partially distributed as well as power efficient intelligence or learning at the network edge (edge cloud). These research trends will further accelerate to adoption of AI/ML in a fully distributed way in 5G, where learning will take place at the edge devices. The trends of processing of data at the

edge will evolve with more intelligence in network having a greater degree of privacy, better efficiency and performance improvement.

Research Trends for solving challenges in 5G

To address various challenges in wireless communication and networks the Telcos, are focusing on the design and development of data-driven solutions controlled by AI/ML. Moreover, there is a trend of the Telco applying ML, learning in the CORE and the RAN [144]. Furthermore, there are research trends to develop intelligent beamforming and efficient power management. As spectrum utilization and the minimization of interference are design issues of prime importance, the Telcos are working on developing such dynamic-spectrum utilization and interference minimizations. The ML is applied also for network system protection to avoid malicious attacks.

ML Incorporation: Various Open Issues Offer Potential Research Directions

The incorporation of ML in the NGWN draws significant attention to several open issues and as well as it attracts multi-dimensional research challenges to combat.

(i) *Time Convergence* The convergence time for run-time decisions, network parameter estimation and the network driving factors have an impact. These are the potential areas of research [22] and development. It is a concern to optimize the time convergence, given the constraint that the convergence due to the ML time convergence must not downgrade the performance of the heavily loaded dynamic network [145].

(ii) *CSI Acquisition in IRS* In an Intelligent Reflecting Surface (IRS) assisted Multiple Input Single Output (MISO) system and in IRS-enhanced wireless communication, it is significantly important that the acquisition of channel State Information (CSI) is in the correct form. Moreover, it must be captured at run-time. In such networks, finding out the CSI is a challenging task due to overhead during the training. Furthermore, in NOMA networks (IRS-guided), in general, the CSI of the users in a cluster is being shared with other users in the same cluster. In such cases, the CSI acquisition process and swap over of information of CSI, are considerably challenging tasks. Moreover, in various cases beyond linear correlations, there are several challenges to deriving the CSI precisely [146]. In such cases, there are research trends that ML and DL-enabled models are being applied to overcome the problems.

(iii) *Resource allocation* AI/ ML can be integrated into various applications related to e-health. In remote medical

treatments using wearable sensors, the challenge is to balance the resource allocation. In case wearable sensors are used for remote medical treatments, there is a research challenge to balance the resource allocation for several technologies. The ML can address such challenges [147].

(iv) *QoS and QoE* Due to the heterogeneity of a network having a large number of users, there are different QoS and QoE and requirements. In video streaming applications, the users expect the least delay but higher throughput. In this case, security is not the utmost concern, although it matters. On the contrary, in the payment software/ gateway, the users demand the highest level of security where the throughput is not the foremost issue. Considering these factors it is a critically challenging task to design the ML protocols applicable to advanced 5G network cross-layer optimizations, network resource utilization and applications [148].

(v) *UaaS* UAVs as an Intelligent Service (UaaS) [22] utilize the technology of UAV to proffer services related to 5G/B5G integrated edge computing [149], edge caching, wireless communication, etc. In UAV ML techniques can be applied for performance improvement. As addressing the proper utilization and the scarcity of the resource is a concern, it is critically essential to apply appropriate models (energy-efficient ML) for training, learning and appropriate decisions for UaaS [22]. Moreover, whenever a UAV works as a trainer for edge intelligence, the system must be designed in such a way that all the participants in the system must undergo energy-efficient training along with the UAV

by applying moderately restricted energy [150]. This is a key research area for further development.

Trends Towards ML-Powered 6G Network: Massive Edge Intelligence

Researchers are working on how 6G will apply massive intelligence incorporating machine learning and mainly the deployment of edge intelligent computing [149]. This means the integration of ML at the network edge which is a challenging task for implementation and draws significant attention. The integration of machine learning in the 6G [150] will offer an optimized network with an improved performance. The following Fig. 12 illustrates the role of ML (target incorporation) in cross-layer optimization of future 6G.

The edge intelligence will enable the network server (centralized or distributed) nearer to the end-users [16] with the support of a cloud network. Massive edge computing applying ML in promising 6G network to be deployed in the near future [151]. In such a network, the users have a closer access to the server from the cloud. The key advantages of incorporating edge computing are [151] (i) increased speed with minimized latency, (ii) improved traffic management, (iii) improved reliability, (iv) and offering enhanced security. With the support of extreme intelligence, there are research trends for the development of intelligent radio access interfaces in THz communication. Performance improvement of the technologies such as; NVF (Network Functions Virtualization), SDN (Software-Defined Networking) and network

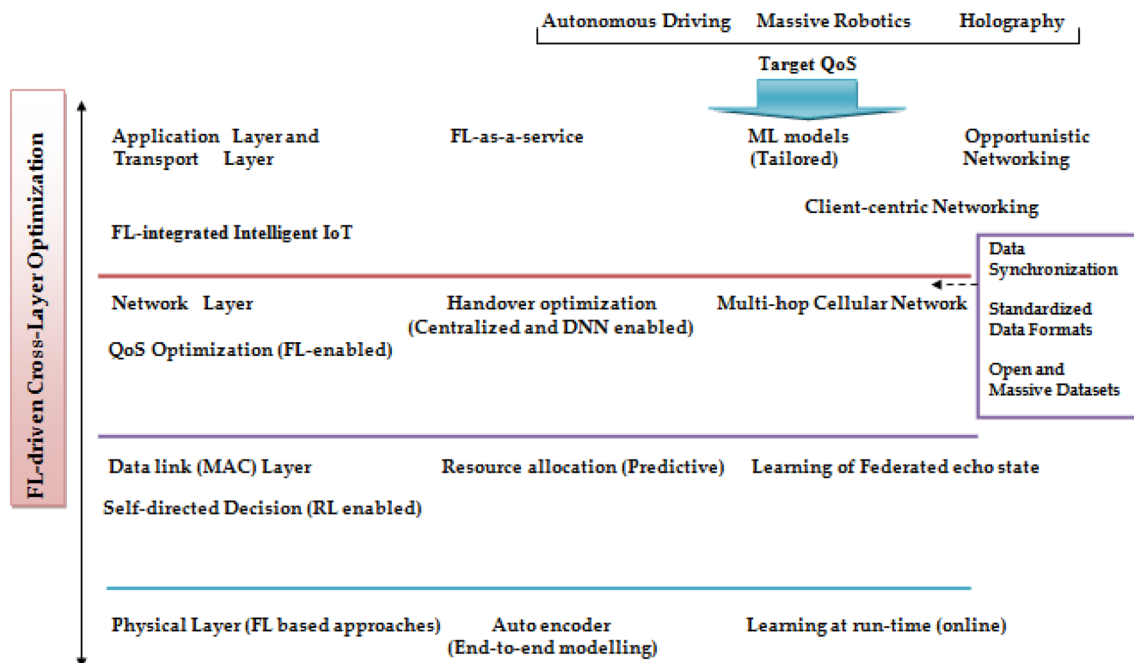


Fig. 12 ML-driven Cross-layer optimization (target incorporation) in 6G

slicing [152], in 6G will be enhanced by applying ML. This will enable the fastest learning and prompt network environment adaption.

The following Table 7 outlines the open research issues and questions to be addressed for the incorporation and development of massive edge intelligence integrating AI/ML techniques in the forthcoming 6G network.

Moreover, the major thrust areas of research trend towards the 6G, is to design and implement trustworthy, reliable, low power, low latency [153], and edge intelligent network. The potential challenges in applying intelligent services are due to the issues of scalability and

trustworthiness with the integration of ML techniques in future 6G [150]. The term ‘trustworthiness’ [16] is deeply related to privacy and as well as security, which are significant concerns for the various services of 6G. Recently, various models of edge learning have been proposed considering the industrial research trends [154] along with the implementation of [155] low latency, high reliability (0.0001% packet loss) communication systems and ML approaches. The typical example is federated learning (FL), [156, 157] as defined in IEEE STD 3652.1–2020 [158] or collaborative learning, split learning [159, 160], swarm learning, trustworthy learning, distributed

Table 7 Open research issues / questions in Edge Intelligence (EI)

Components and Implementing Factors of Edge Intelligence	Open and Key Research Issues/ Questions to be focused
Edge Infrastructure	1. The considerations of the architectural issues for various developments of the applications of edge intelligence
Required Platform	1. The practices of software development and testing methods to be deployed in edge intelligence-based applications 2. The Development and Operations (DevOps) features 3. The involvement of several edge units running in a decentralized setting 4. The incorporation of the methods (processes and tools) of formal verification
Network management	1. To address learning (online) of running data or applications of federated learning in an energy- efficient way 2. The issues of distributed computing among heterogeneous resources
Incorporation of Edge-device Intelligence	1. For the integration of the solutions of AI (lightweight) at the edge 2. The ML/AI based techniques to address horizontal and as well as vertical collaborative and cooperative computing including the optimized genetic algorithms, game theory and swarm intelligence- based approaches
Incorporation of required software at the Edge	1. The interoperability and flexibility of the software agents with the quality of re-configurability, supporting virtualization 2. The deployment and flow of the liquid software
Online learning and real-time operations	1. The co-design issues related to the communication and control along with the ML to be deployed 2. The prompt adaptability of edge-intelligent model related to training and to the rapid changes of network environment-dynamics 3. The issues of processing and computational complexities
Training (distributed type), Algorithm Design and Integration	1. The role of accelerators in a distributed training and ML design approaches 2. The deployment of sufficient resources (hardware) at the edge for ML in a cost-effectively manner 3. The implementation of energy efficient computation, minimized power consumption and optimized storage at the edge 4. The running of the training and ML algorithm in part in heterogeneous platform and estimation of the network environment parameters at run-time
Security and privacy issues	1. Data security, protection of the ML model-tampering 2. The impact of operations during a node/ nodes or in a worst case of the network fault/ failure 3. Users’ privacy concern, the compliance of the privacy of with the regulatory bodies
Concerns of the End users	1. The issues of mobility and resource sharing 2. The various issues to be addressed for the standardization of the KPI associated with the QoE 3. The impact of the edge-intelligence particularly the social and technological impact
Pricing methods	1. The virtualization of communication along with caching and computation with the learning of the resources 2. The mechanism of dynamic pricing and revenue generation

reinforcement learning (DRL) [161, 162] to deal with the privacy issues and adversarial attacks in NGWN [154].

FL Addressing Forthcoming 6G

The deployment of Federated Learning (FL) for edge intelligence in 6G will invite a revolution in the telecom sector. The key performing metrics of FL are delay, connectivity, reliability and energy, which are highlighted as follows:

FL-Delay The delay includes the delay due to devices (at the edge), uplink and downlink delays and for the BS aggregation. The time due to the number of iterations in FL required to converge also has a key role [135] in deciding the system delay. The challenge is to minimize the overall FL delay balancing the delays of communication and computing.

FL-Connectivity Due to the massive connectivity requirements for a good number of devices, it is a challenging task to implement by using conventional channel access mechanisms as they suffer due to significant delay. To combat such challenges, a computation approach of over-the-air can be applied in a superposition way in wireless transmission [164, 165]. Furthermore, applying FL for the scheduling of a small part of devices at every turn is an alternate promising approach [166, 167] for network performance optimization.

FL-Reliability Due to resource constraints (for example bandwidth) and unreliability of the communication links, significant training errors may outcome. The symbol errors due to the untrustworthy wireless channels and the constraint of the availability of resources may affect the FL performance as well as the FL iterations (success rate) [168]. In general, these factors affect the speed of convergence of FL and the overall system's performance.

FL-Energy The training and execution of FL are affected by the computational energy (local) and the transmission energy of the wireless node/ devices. The computation energy of a node/ device is related to the number of iterations required for the computation (local). The transmission

energy is decided by the count of iterations during the operations of the FL.

FL has the potential attributes to address the 6G network characteristics:

mURLLC Due to the exponential growth in number of the end-user devices, the various metrics and QoS parameters of 5G Ultra-reliable low latency communications (URLLC) must be upgraded to Massive ultra-reliable low latency communications (mURLLC) in such a way that multiple computing (edge) units can be utilized to learn from the network (shared) in a cooperative manner, resulting minimization of the delay (service) and offers higher reliability [169].

Scalable-Architecture The target implementation of FL includes the implementation of edge intelligence in a distributed way which offers the services of a large number of devices/ end users to be connected with edge servers. Applying FL, the scalable and decomposable architecture provides the advantages of concurrent computing of several edge servers.

Offered services (Human-centric) As 6G will offer various services (human-centric), which will need significant quality of experience associated with the physical movements of the end users, where FL can easily predict the end user's gestures along with the results (predicted) of the base stations to offer a better QoE.

The following Table 8 summarizes the various contemporary contributions related to the approaches of FL in edge intelligence in the 6G network.

In addition to the aforementioned studies related to the pros and cons and various state-of-the-art approaches of the FL, several challenges arise due to computational complexity, storage and scalability. Moreover, various challenges are energy consumption, edge latencies (consisting of computation along with communication latency) and expected accuracy of edge intelligence deployment. The present research trends demand the transition of a new age of wireless system design from 'data-oriented' communication to the rolling out towards 'task-oriented' communication. The data-oriented communication signifies the reliability or the

Table 8 The approaches of FL in edge computing for 6G

Reference	FL Approaches/ Methods	Areas / Contributions
[170]	FL and RL	Optimization of the edge computing (mobile) and for requirements of caching
[171]	FL	FL integration and challenges
[172]	Edge-ML	Illustrations of the architecture of Edge-ML
[173]	FL	Integration and characteristics of FL
[174]	FL	Use cases/ Application and Services of FL
[175]	Edge-ML	FL incorporation challenges in the systems of the edge of the network
[176]	FL	FL Challenges, performance metrics, implementation challenges in 6G
[177]	Edge Computing, FL	Edge computing, challenges, use cases, market survey
[178]	FL	FL Methods, incorporation challenges, future trends in 6G

increasing data rate of the communication system. This is dependent on Shannon's theory [179]. On the contrary, task-oriented communication signifies applying fast intelligence with accuracy at the edge of the network.

The future 6G networks will be extremely complex and dynamic due to the characteristics of its heterogeneity [180] and scale. Moreover, there is a requirement for dense deployment. Therefore, existing methodologies of the wireless network may not be applied for target performance optimization and desired QoS/QoE. For example, in the case of modern traditional problems, the researchers treat the objective function in a conventional algebraic form. As a result, it becomes easy for an optimizer for the requirements of the process of evaluation and infer a result by simple calculation. On the contrary, in future 6G networks, the target mapping among a decision and its outcome in the system is a cost-prohibited one. It may not be obtainable analytically. To address such issues in the desired implementation, reinforcement learning (RL) and deep reinforcement learning (DRL) [181] can be applied as potential solutions. A feedback loop, between, the physical system and the decision-maker can be configured in such a way, that the actions can be iteratively refined by the decision-maker. However, the time spent to arrive at a solution must be taken care of. The time must be significantly minimal to support run-time decisions in the end devices of the edge-intelligent network.

The challenge for the development and implementation of the intelligence at the edge [150], in the 6G could be developed by applying a distributed neural network for network optimization [150]. The trend towards the implementation of intelligence at the edge provides a direction towards advanced network sensing [150], network dynamics and overall smart decisions. Moreover, edge intelligence provides advantageous allocation of resources in a smarter way [182]. Recently, there have been trends to apply decentralized and distributed RL [27, 183], graph neural networks [184], and distributed deep neural network (DNN) [185], for automatic learning at run-time. These are utilized along with the applications of various optimized resource allocation algorithms intelligently.

Conclusion

This paper has portrayed a comprehensive survey on incorporating machine learning for the requirements of intelligence in advanced 5G, beyond (B5G) and towards the forthcoming 6G network. The author has made a comprehensive review of various state-of-the-art ML approaches, and techniques integrating ML for NGWN. The paper illustrates potential target applications of ML in NGWN. In this survey, several challenges due to the deployment of the next generation wireless network and ML incorporation in

the advanced network have been investigated. These challenges are the barriers against the fastest rolling out from advanced 5G to B5G and towards the 6G network. The paper has outlined various open issues which offer potential future research directions. Moreover, the author has highlighted the research trends and pros and cons of emerging network edge intelligence including the integration of promising FL in future 6G.

Acknowledgements The author acknowledges the various telecom industry white papers he has referred to while preparing this manuscript which have been cited in the reference and the opinion of the telecom industry professionals.

Author Contributions The study and the investigations in this paper has been contributed by Dr. Paul as the sole author.

Funding No funding was received for conducting this study.

Data Availability The datasets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Code Availability Not applicable.

Declarations

Conflicts of Interest No conflict of interest exists. The author wishes to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

Human Participants and/or Animals The current study does not contain any studies with human participants and/or animals.

References

1. Special Report: 5G roll-out in the EU: delays in deployment of networks with security issues remaining unresolved, 2022, European Court of Auditors. Available: https://www.eca.europa.eu/Lists/ECADocuments/SR22_03/SR_Security-5G-networks_EN.pdf.
2. 5G Unlocks A World of Opportunities – Top Ten 5G Use Cases”, whitepaper of Huawei, 2017, pp. 1–30. <https://www.huawei.com/en/technology-insights/industry-insights/outlook/mobile-broadband/insights-reports/5g-unlocks-a-world-of-opportunities>.
3. Wikstgrom G, et al. 6G connecting a Cyber Physical World: a Research outlook toward 2030, Ericsson white paper, February, 2022, pp. 3–6, Available: <https://www.ericsson.com/4927de/assets/local/reports-papers/white-papers/6g--connecting-a-cyber-physical-world.pdf>.
4. Where 5G Technology has been deployed , GSA 5G Snapshot on Statista, July, 2022. Available: <https://www.statista.com/chart/23194/5g-networks-deployment-world-map/>.
5. The Global Mobile Suppliers Association (GSA) Report Available: <https://gsacom.com/paper/5g-market-snapshot-june-2022/>.
6. Ericsson Mobility Report, June 2022. <https://www.ericsson.com/49d3a0/assets/local/reports-papers/mobility-report/documents/2022/ericsson-mobility-report-june-2022.pdf>.
7. Gozalvez J. Tentative 3GPP timeline for 5G [mobile radio]. IEEE Vehicular Technol Mag. 2015;10(3):12–8.

8. Akhtar MW, Hassan SA, Ghaffar R, et al. The shift to 6G communications: vision and requirements. *Hum Cent Comput Inf Sci.* 2020;10:53. <https://doi.org/10.1186/s13673-020-00258-2>.
9. Mourad A, Yang R, Lehne PH, De La Oliva A. A baseline roadmap for advanced wireless research beyond 5G. *Electronics.* 2020;9(2):351.
10. Kerravala Z. The success of artificial intelligence and machine learning requires an architectural approach to infrastructure, whitepaper of ZK Research, August 2018, pp. 1–9. https://transformation.cisco.com/c/pxposteventAIandML_en?x=lp2LHG
11. AI and machine Learning: Why now?: Nnetwork optimization in the edge of 5G, Report of Senza Fili, in support with Intel, Nokia and Uhana, 2019. <https://www.intel.com> pp.2–37. <https://www.intel.com/content/dam/www/public/us/en/documents/reports/ai-and-5g-report.pdf>
12. Singh R, Mehbodniya A, Webber JL, Dadheech P, Pavithra G, Alzaidi MS, Akwafo, . Analysis of network slicing for management of 5G networks using machine learning techniques. *Wirel Commun Mob Comput.* 2022. <https://doi.org/10.1155/2022/9169568>.
13. Ali S, Saad W, Steinbach D. 6G flagship program, University of Oulu, Finland. White Paper on Machine Learning in 6G Wireless Communication Networks, June 2020, pp. 1–34, 6G Research Visions, No. 7, ISBN 978–952–62–2673–6. <http://jultika.oulu.fi/files/isbn9789526226736.pdf>.
14. Berardinelli G, et al. Extreme communication in 6G: vision and challenges for ‘in-X’ Subnetworks. *IEEE Open J Commun Soc.* 2021;2:2516–35. <https://doi.org/10.1109/OJCOMS.2021.3121530>.
15. Nawaz SJ, Sharma SK, Wyne S, Patwary MN, Asaduzzaman M. Quantum machine learning for 6G communication networks: state-of-the-art and vision for the future. *IEEE Access.* 2019;7:46317–50. <https://doi.org/10.1109/ACCESS.2019.2909490>.
16. 6 G : The Next Hyper-Connected Experience for All, -Samsung 6G Vision, white paper of Samsung Research, July, 2020. pp. 19–20. <https://research.samsung.com/>.
17. A Vision of Artificial Intelligence for 6G Communications, A Thought Leadership White Paper. August, 2022, Comarch, pp. 1–13. <https://www.comarch.com>.
18. 5G Evolution and 6G, Whitepaper of NTT DOCOMO, INC. (Ver. 4), January 2022, pp. 1–60. https://www.docomo.ne.jp/english/binary/pdf/corporate/technology/whitepaper_6g/DOCOMO_6G_White_PaperEN_v4.0.pdf.
19. Zhang Z, et al. 6G wireless networks: Vision, requirements, architecture, and key technologies. *IEEE Veh Technol Mag.* 2019;14(3):28–41. <https://doi.org/10.1109/MVT.2019.2921208>.
20. Nayak S, Patgiri R. 6G communication technology: A vision on intelligent healthcare. 2020; pp. 1–9. <https://doi.org/10.48550/arXiv.2005.07532>.
21. Alwis CD, Kalla A, Pham Q-V, Kumar P, Dev K, Hwang W-J, Liyange M. Survey on 6G frontiers: trends, applications, requirements, technologies and future research. *IEEE Open J Commun Soc.* 2021. <https://doi.org/10.1109/OJCOMS.2021.3071496>.
22. Rekkas VP, Sotiroudis S, Sarigiannidis P, Wan S, Karagiannidis GK, Goudos SK. Machine learning in beyond 5G/6G networks-state-of-the-art and future trends. *Electronics.* 2021;10:1–28. <https://doi.org/10.3390/electronics10222786>.
23. Chang L, Zhang Z, Li P, Xi S, Guo W, Shen Y, Xiong Z, Kang J, Niyato D, Qiao X, Wu Y. 6G-Enabled Edge AI for metaverse: challenges, methods, and future research directions. *J Commun Inform Netw.* 2022; 7(2):107–121. <https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9815195>.
24. Deng L, Liu K, Zhang Y, Banzi M, Casey S, Cohen B. White paper: intelligent networking, AI and machine learning—a telecommunications operator’s perspective, LF Networking. 2022; pp. 1–26. https://lfnetworking.org/wp-content/uploads/sites/7/2022/06/LFN_IntelligentNetworking_Whitepaper_110421.pdf.
25. Ignite 5G Network Analytics with Machine Learning & Artificial Intelligence, Whitepaper of Fujitsu Network Communications, Inc., 2022, pp.1–6. https://fujitsu.lookbookhq.com/ignite22?utm_source=blog&utm_medium=fujitsu-site.
26. Corcoran D, Ermedahl A, Granbom C. Artificial Intelligence in RAN : A Software framework for AI-Driven RAN Automation. *Ericsson Technology Review*, 2020; December 8, 2020, pp. 1–12, ISSN: 0014–0171. <https://www.ericsson.com/4ae5c4/assets/local/reports-papers/ericsson-technology-review/docs/2020/artificial-intelligence-in-ran.pdf>.
27. Intelligent Operations: How AI plays a critical role in Network operations, Intelligent Automation Guide Series, (2022) Ericsson white paper, 0341–8/22112-FGM1010 Uen, June 2022, pp.2–8. <https://www.ericsson.com/49d443/assets/local/core-network/doc/intelligent-operations-guide.pdf>.
28. The Nokia AVA Telco AI and analytics solutions. <https://www.nokia.com/networks/ai-and-analytics/#what-is-AVA-telecom-analytics>.
29. 5G AND Beyond Technology Roadmap, IEEE whitepaper on IEEE. pp. 1–33. <https://futurenetworks.ieee.org/images/files/pdf/ieee-5g-roadmap-white-paper.pdf>.
30. Beyond 5G/6G (2022), whitepaper of NICT: National institute of Information and Communication Technology, Beyond 5G promotion unit, Japan, June 2022, pp. 6–11. https://beyond5g.nict.go.jp/images/download/NICT_B5G6G_WhitePaperEN_v2_0.pdf.
31. Employing AI techniques to enhance returns on 5G network investments, Ericsson Report, (Ericsson AI and Automation), pp. 1–12, BNEW-19:007926 Uen. <https://www.ericsson.com/49b63f/assets/local/ai-and-automation/docs/machine-learning-and-ai-aw-screen.pdf>.
32. Ssengonzi C, Kogeda OP, Olwal T O (2022) A survey of deep reinforcement learning application in 5G and beyond network slicing and virtualization. *Array.* 2022;14:1–27. <https://doi.org/10.1016/j.array.2022.100142>.
33. Tong W, Zhu P. Next Horizons- from connected People and things to connected Intelligence. Huawei Communications Technologies Labs, 6G, Cambridge University Press; 2021. <https://doi.org/10.1017/9781108989817>.
34. Adib D. The impact of 5G on the manufacturing industry: A \$740BN Opportunity, A report of STL Partners with contribution from Huawei, October 201, pp. 1–40. 2019. <https://carrier.huawei.com/~media/CNBNV2/download/program/Industries-5G/5G-Impact-on-Manufactureing.pdf>.
35. Jiang C, Zhang H, Ren Y, Han Z, Chen K-C, Hanzo L. Machine learning paradigms for next-generation wireless networks. *IEEE Wirel Commun.* 2017;24(2):98–105. <https://doi.org/10.1109/MWC.2016.1500356WC>.
36. Salameh AI, Tarhuni ME. From 5G to 6G—challenges technologies, and applications. *Future Internet.* 2022;14(117):1–35. <https://doi.org/10.3390/fi14040117>.
37. Akhtar MW, Hassan SA, Ghaffar R, Jung H, Garg S, Hossain MS. The shift to 6G communications: vision and requirements. *Springer Human-Centric Comput Inform Sci.* 2020;53(10):1–27. <https://doi.org/10.1186/s13673-020-00258-2>.
38. Ly A, Yao Y-D. A review of deep learning in 5G research: channel coding, massive MIMO, multiple access, resource allocation, and network security. *IEEE Open Journal of the Commun Soc.* 2021;2:396–408. <https://doi.org/10.1109/OJCOMS.2021.3058353>.
39. Samanta RK, et al. Scope of machine learning applications for addressing the challenges in next-generation wireless networks. *CAAI Trans Intell Technol.* 2022;7(3):395–418. <https://doi.org/10.1049/cit2.12114>.

40. Iliadis LA, Zaharis ZD, Sotiroidis S, Sarigiannidis P, Karagianidis GK, Goudos SK. The road to 6G: a comprehensive survey of deep learning applications in cell-free massive MIMO communications systems. *EURASIP J Wirel Commun Netw*. 2022;68:1–16. <https://doi.org/10.1186/s13638-022-02153-z>.
41. Mahmood MR, Matin MA, Sarigiannidis P, Goudos SK. A Comprehensive review on artificial intelligence/machine learning algorithms for empowering the future IoT toward 6G Era. *IEEE Access*. 2022;10:87535–62. <https://doi.org/10.1109/ACCESS.2022.3199689>.
42. Liu Y, Deng Y, Nallanathan A, Yuan J. Machine learning for 6G enhanced ultra-reliable and low-latency services. *IEEE Wirel Commun*. 2023;30(2):48–54. <https://doi.org/10.1109/MWC.006.2200407>.
43. Ferrag MA, et al. Edge learning for 6G-enabled Internet of Things: a comprehensive survey of vulnerabilities, datasets, and defenses. *IEEE Commun Surv Tutor*. 2023;25(4):2654–713. <https://doi.org/10.1109/COMST.2023.3317242>.
44. Bárcena JLC, et al. Enabling federated learning of explainable AI models within beyond-5G/6G networks. *Comput Commun*. 2023;210(1):356–75. <https://doi.org/10.1016/j.comcom.2023.07.039>.
45. de Figueiredo FAP. An overview of massive MIMO for 5G and 6G. *IEEE Latin Am Trans*. 2022;20(6):931–40. <https://doi.org/10.1109/TLA.2022.9757375>.
46. Falahy NA, Alani OY. Technologies for 5G networks: challenges and opportunities, IT Pro, IEEE, 2017; pp. 12–20. <https://doi.org/10.1109/MITP.2017.9>
47. Carlowitz C, Dietz M. Integrated front-end approaches for wireless 100 Gb/s and beyond: enabling efficient ultra-high speed wireless communication systems. *IEEE Microwave Mag*. 2023;24(8):16–34. <https://doi.org/10.1109/MMM.2023.3277360>.
48. Huang J, Ruan F, Su M, Yang X, Yao S, Zhang J. Analysis of orthogonal frequency division multiplexing (OFDM) technology in wireless communication process. In: 2016 10th IEEE International Conference on Anti-counterfeiting, security, and identification (ASID), Xiamen, 2016, pp. 122–125. <https://doi.org/10.1109/ICASID.2016.7873931>.
49. Rao L, Pant M, Malviya L, Parmar A, Charhate SV. 5G beam-forming techniques for the coverage of intended directions in modern wireless communication: in-depth review. *Int J Microw Wirel Technol*. 2020;13(10):1039–62. <https://doi.org/10.1631/FITEE.1601817>.
50. Papidas AG, Polyzos GC. Self-organizing networks for 5G and beyond: a view from the top. *Future Internet*. 2022;14(3):1–30. <https://doi.org/10.3390/fi14030095>.
51. ITU-FG ML 5G focus group. <https://www.itu.int/en/ITU-T/focusgroups/ml5g/Pages/default.aspx>.
52. Zhang Z, Xiao Y, Ma Z, Xiao M, Ding Z, Lei X, Karagianidis GK, Fan P. 6G wireless networks: vision, requirements, architecture, and key technologies. *IEEE Veh Technol Mag*. 2019;14(3):28–41. <https://doi.org/10.1109/MVT.2019.2921208>.
53. Guan W, Zhang H, Leung VC. Customized slicing for 6G: enforcing artificial intelligence on resource management. *IEEE Network*. 2021;35(5):264–71. <https://doi.org/10.1109/MNET.011.2000644>.
54. Xie F, Wei D, Wang Z. Traffic analysis for 5G network slice based on machine learning. *EURASIP J Wirel Commun Netw*. 2021;108:1–15. <https://doi.org/10.1186/s13638-021-01991-7>.
55. AI and ML—Enablers for Beyond 5G Networks (2021), white-paper of 5G PPP Technology Board of 5G PPP (A joint initiative between the European Commission and European ICT industry, ICT manufacturers, telecommunications operators, service providers, SMEs and researcher Institutions), May 2021, pp.10-22. 10.5281/zenodo.4299895.
56. Ericsson Mobility Report, June 2023. <https://www.ericsson.com/49dd9d/assets/local/reports-papers/mobility-report/documents/2023/ericsson-mobility-report-june-2023.pdf>.
57. 5 G network slicing and self-management, white paper of 5G slicing Association, pp. 2–25, October, 2020. <https://www-file.huawei.com/-/media/corporate/pdf/news/5g-network-slicing-self-management-white-paper.pdf?la=en-us>.
58. Artificial Intelligence Index Report 2023, HAI, Stanford University, pp.1–386, 2023. https://aiindex.stanford.edu/wp-content/uploads/2023/04/HAI_AI-Index-Report_2023.pdf.
59. Khwandah SA, Cosmas JP, Lazaridis PI, et al. Massive MIMO systems for 5G communications. *Wireless Pers Commun*. 2021;120:2101–15. <https://doi.org/10.1007/s11277-021-08550-9>.
60. Astely D, Butovitsch PV, Faxer S, Larsson E. Meeting 5G network requirements with Massive MIMO, Ericsson Technology Review, pp. 2–10, February 16, 2022, ISSN 0014–0171. <https://www.ericsson.com/4917a1/assets/local/reports-papers/ericsson-technology-review/docs/2022/the-role-of-massive-mimo-in-5g-networks.pdf>.
61. Michalski RS, Carbonell JG, Mitchell TM. mMachine learning: an artificial intelligence approach. Springer Science & Business Media; 2013. <https://doi.org/10.1007/978-3-662-12405-5>.
62. Safi H, Montazeri AM, Rostampoor J, Parsaeefard S. Spectrum sensing and resource allocation for 5G heterogeneous cloud radio access networks. *IET Commun*. 2022;16(4):348–58. <https://doi.org/10.1049/cmu2.12356>.
63. Ruan L, Dias MPI, Wong E. Machine learning-based bandwidth prediction for low-latency H2M applications. *IEEE Internet Things J*. 2019;6(2):3743–52. <https://doi.org/10.1109/JIOT.2018.2890563>.
64. Yang G, Zhang Y, He Z, Wen J, Ji Z, Li Y. Machine-learning-based prediction methods for path loss and delay spread in air-to-ground millimetre-wave channels. *IET Microwaves Antennas Propag*. 2019;13(8):1113–21. <https://doi.org/10.1049/iet-map.2018.6187>.
65. Yuan J, Ngo HQ, Matthaiou M. Machine learning-based channel prediction in massive MIMO with channel aging. *IEEE Trans Wirel Commun*. 2020;19(9):2960–73. <https://doi.org/10.1109/TWC.2020.2969627>.
66. Alrabeiah M, Alkhateeb A. Deep Learning for TDD and FDD massive MIMO: mapping channels in space and frequency, In. *Proc. 2019 53rd Asilomar Conference on signals, systems, and computers*, 2019; pp. 1465–1470. <https://doi.org/10.1109/IEEECONF44664.2019.9048929>.
67. Chen M, Saad W, Yin C. Liquid state machine learning for resource and cache management in LTE-U unmanned aerial vehicle (UAV) networks. *IEEE Trans Wirel Commun*. 2019;18(3):1504–17. <https://doi.org/10.1109/TWC.2019.2891629>.
68. Antón-Haro C, Mestre X. Learning and data-driven beam selection for mmWave communications: an angle of arrival-based approach. *IEEE Access*. 2019;7:20404–15. <https://doi.org/10.1109/ACCESS.2019.2895594>.
69. Nadig D, Ramamurthy B, Bockelman B., Swanson D. APRIL: an application-aware, predictive and intelligent load balancing solution for data-intensive science. In: *Proc. IEEE INFOCOM 2019-IEEE Conf. on Computer Communications*, Paris, 29 April–2 May 2019; pp. 1909–1917. <https://doi.org/10.1109/INFOCOM.2019.8737537>.
70. Abuzainab N, Alrabeiah M, Alkhateeb A, Sagduyu YE. Deep learning for THz drones with flying intelligent surfaces: beam and handoff prediction. In: *Proc. IEEE International Conference on Communications Workshops*, Canada, 14–23 June 2021, pp. 1–6. <https://doi.org/10.1109/ICCWorkshops50388.2021.9473804>.

71. Wang W, Zhu M, Wang J, Zeng X, Yang Z. End-to-end encrypted traffic classification with one-dimensional convolution neural networks. In: Proc. IEEE International Conf. on Intelligence and Security Informatics (ISI), Beijing, China, 22–24 July 2017; pp. 43–48. <https://doi.org/10.1109/ISI.2017.8004872>.
72. Kwon HJ, Lee JH, Choi W. Machine learning-based beamforming in K-user MISO interference channels. IEEE Access. 2021;9:28066–75. <https://doi.org/10.1109/ACCESS.2021.3058759>.
73. Kim J, Choi JP. Sensing coverage-based cooperative spectrum detection in cognitive radio networks. IEEE Sens J. 2019;19(13):5325–32. <https://doi.org/10.1109/JSEN.2019.2903408>.
74. Yang Y, Gao Z, Ma Y, Cao B, He D. Machine learning enabling analog beam selection for concurrent transmissions in millimeter-wave V2V communications. IEEE Trans Veh Technol. 2020;69(8):9185–9. <https://doi.org/10.1109/TVT.2020.3001340>.
75. Sliwa B, Adam R, Wietfeld C. Client-based intelligence for resource efficient vehicular big data transfer in future 6G networks. IEEE Trans Veh Technol. 2021;70(6):5332–46. <https://doi.org/10.1109/TVT.2021.3060459>.
76. Sliwa B, Falkenberg R, Wietfeld C. Towards cooperative data rate prediction for future mobile and vehicular 6G networks. In: Proc. 2nd 6G Wireless Summit (6G SUMMIT), Virtual, 17–20 March 2020; pp. 1–5. <https://doi.org/10.1109/6GSUMMIT49458.2020.9083767>.
77. Wei Y, Yu FR, Song M, Han Z. Joint optimization of caching, computing, and radio resources for fog-enabled IoT using natural actor-critic deep reinforcement learning. IEEE Internet Things J. 2018;6(2):2061–73. <https://doi.org/10.1109/JIOT.2018.2878435>.
78. Chu M, Li H, Liao X, Cui S. Reinforcement learning-based multi access control and battery prediction with energy harvesting in IoT systems. IEEE Internet Things J. 2019;6(2):2009–20. <https://doi.org/10.1109/JIOT.2018.2872440>.
79. Manesh MR, Kenney J, Hu WC, Devabhaktuni, V.K.; Kaabouch, N. Detection of GPS spoofing attacks on unmanned aerial systems. In Proc. 16th IEEE Annual Consumer Communications & Networking Conference (CCNC), Las Vegas, USA, 11–14 January 2019; pp. 1–6. <https://doi.org/10.1109/CCNC.2019.8651804>.
80. Goudos SK, Tsoulos GV, Athanasiadou G, Batistatos MC, Zarbouti D, Psannis KE. Artificial neural network optimal modeling and optimization of UAV measurements for mobile communications using the L-SHADE algorithm. IEEE Trans Antennas Propag. 2019;67(6):4022–31. <https://doi.org/10.1109/TAP.2019.2905665>.
81. Goudos SK, Athanasiadou G. Application of an ensemble method to UAV power modeling for cellular communications. IEEE Antennas Wirel Propag Lett. 2019;18(11):2340–4. <https://doi.org/10.1109/LAWP.2019.2926784>.
82. Si H, Wang Y, Yuan J, Shan X. Mobility prediction in cellular network using hidden markov model. In Proc. 7th IEEE Consumer Communications and Networking Conference, Las Vegas, NV, USA, 9–12 January 2010; pp. 1–5. <https://doi.org/10.1109/CCNC.2010.5421684>.
83. Mohamed A, Onireti O, Hoseinitabatabaei SA, Imran M, Imran A, Tafazolli R. (2015) Mobility prediction for handover management in cellular networks with control/data separation. In: Proc. IEEE International Conference on Communications (ICC), London, UK, 8–12 June 2015; pp. 3939–3944. <https://doi.org/10.1109/ICC.2015.7248939>.
84. Hassan N, Hossan MT, Tabassum H. User association in coexisting RF and tera hertz networks in 6G. In Proc. IEEE Canadian Conference on Electrical and Computer Engineering (CCECE), London, ON, Canada, 30 August–2 September 2020; pp. 1–5. <https://doi.org/10.1109/CCECE47787.2020.9255737>.
85. Rajendran S, Meert W, Giustiniano D, Lenders V, Pollin S. Deep learning models for wireless signal classification with distributed low-cost spectrum sensors. IEEE Trans Cognit Commun Netw. 2018;4(3):433–45. <https://doi.org/10.1109/TCCN.2018.2835460>.
86. West NE, O'Shea T. Deep architectures for modulation recognition. In: Proceedings of the 2017 IEEE International Symposium on Dynamic Spectrum Access Networks (DySPAN), Baltimore, MD, USA, 6–9 March 2017; pp. 1–6. <https://doi.org/10.1109/DySPAN.2017.7920754>.
87. Gómez-Andrades A, Munoz P, Serrano I, Barco R. Automatic root cause analysis for LTE networks based on unsupervised techniques. IEEE Trans Veh Technol. 2016;65(4):2369–86. <https://doi.org/10.1109/TVT.2015.2431742>.
88. Hashmi US, Darbandi A, Imran A. Enabling proactive self-healing by data mining network failure logs. In Proc. International Conference on computing, networking and communications (ICNC), Silicon Valley, CA, USA, 26–29 January 2017; pp. 511–517. <https://doi.org/10.1109/ICCNC.2017.7876181>.
89. Samuel N, Diskin T, Wiesel A. Deep MIMO detection. In: Proc. IEEE 18th International Workshop on Signal Processing Advances in Wireless Communications (SPAWC), Sapporo, Japan, 3–6 July 2017; pp. 1–5. <https://doi.org/10.1109/SPAWC.2017.8227772>.
90. Fan Z, Gu X, Nie S, Chen M. D2D power control based on supervised and unsupervised learning. In: Proc. 3rd IEEE International Conference on computer and communications (ICCC), Chengdu, China, 13–16 December 2017; pp. 558–563. <https://doi.org/10.1109/CompComm.2017.8322607>.
91. Sattiraju R, Weinand, A, Schotten HD. AI-assisted PHY technologies for 6G and beyond wireless networks. In: 1st 6G Wireless Summit, March 24–25, 2019, Levi, Finland pp. 1–2. <https://doi.org/10.48550/arXiv.1908.09523>.
92. Liu Y, Qin Z, Cai Y, Gao Y, Li GY, Nallanathan, A. A UAV communications based on non-orthogonal multiple access. IEEE Wirel Commun. 2019;26(1):52–7. <https://doi.org/10.1109/MWC.2018.1800196>.
93. Xiao L, Wan X, Lu X, Zhang Y, Wu D. IoT security techniques based on machine learning: How do IoT devices use AI to enhance security? IEEE Signal Process Mag. 2018;35(5):41–9. <https://doi.org/10.1109/MSP.2018.2825478>.
94. Yu Y, Long J, Cai Z. Network intrusion detection through stacking dilated convolutional autoencoders. Securit Commun Netw. 2017;4184196:1–10. <https://doi.org/10.1155/2017/4184196>.
95. Cui J, Ding Z, Fan P, Al-Dhahir N. Unsupervised machine learning-based user clustering in millimeter-wave-NOMA systems. IEEE Trans Wirel Commun. 2018;17(11):7425–40. <https://doi.org/10.1109/TWC.2018.2867180>.
96. Ren J, Wang Z, Xu M, Fang F, Ding Z. An EM-based user clustering method in non-orthogonal multiple access. IEEE Trans Commun. 2019;67(12):8422–34. <https://doi.org/10.1109/TCOMM.2019.2945334>.
97. Mohamed A, Ruan H, Abdelwahab MHH, Dorneanu B, Xiao P, Arellano-Garcia H, Gao Y, Tafazolli R. An interdisciplinary modelling approach in industrial 5G/6G and machine learning era. In: Proc. IEEE International Conference on Communications Workshops (ICC Workshops), Dublin, Ireland, 7–11 June 2020; pp. 1–6. <https://openresearch.surrey.ac.uk/esploro/outputs/99511929802346>.
98. Chen Y, Zhang Y, Maharjan S, Alam M, Wu T. Deep learning for secure mobile edge computing in cyber-physical transportation systems. IEEE Network. 2019;33(4):36–41. <https://doi.org/10.1109/MNET.2019.1800458>.
99. Huang H, Xia W, Xiong J, Yang J, Zheng G, Zhu X. Unsupervised learning-based fast beamforming design for downlink MIMO. IEEE Access. 2018;7:7599–605. <https://doi.org/10.1109/ACCESS.2018.2887308>.

100. Xu X, Zuo L, Huang Z. Reinforcement learning algorithms with function approximation: recent advances and applications. *Inf Sci.* 2014;261:1–31. <https://doi.org/10.1016/j.ins.2013.08.037>.
101. Tanveer J, Haider A, Ali R, Kim A. An overview of reinforcement learning algorithms for handover management in 5G ultra-dense small cell networks. *Appl Sci.* 2022;12(1):1–25. <https://doi.org/10.3390/app12010426>.
102. Azimi Y, Yousefi S, Kalbkhani H, Kunz T. Energy-efficient deep reinforcement learning assisted resource allocation for 5G-RAN slicing. *IEEE Trans Veh Technol.* 2022;71(1):856–71. <https://doi.org/10.1109/TVT.2021.3128513>.
103. Tang J, Tang H, Zhang X, Cumanan K, Chen G, Wong KK, Chambers JA. Energy minimization in D2D-assisted cache-enabled Internet of Things: a deep reinforcement learning approach. *IEEE Trans Industr Inf.* 2020;16(8):5412–23. <https://doi.org/10.1109/TII.2019.2954127>.
104. Sun Y, Peng M, Mao S. Deep reinforcement learning-based mode selection and resource management for green fog radio access networks. *IEEE Internet Things J.* 2018;6(2):1960–71. <https://doi.org/10.1109/JIOT.2018.2871020>.
105. Li Z, Guo C, Xuan Y. A multi-agent deep reinforcement learning based spectrum allocation framework for D2D communications. In: *Proc. IEEE Global Communications Conference (GLOBECOM)*, Waikoloa, HI, USA, 9–13 December, 2019; pp. 1–6. <https://doi.org/10.1109/GLOBECOM38437.2019.9013763>.
106. Shah HA, Zhao L, Kim IM. Joint network control and resource allocation for space-terrestrial integrated network through hierarchical deep actor-critic reinforcement learning. *IEEE Trans Veh Technol.* 2021;70(5):4943–54. <https://doi.org/10.1109/TVT.2021.3071983>.
107. Kang JM. Reinforcement learning based adaptive resource allocation for wireless powered communication systems. *IEEE Commun Lett.* 2020;24(8):1752–6. <https://doi.org/10.1109/LCOMM.2020.2988817>.
108. Mollel MS, Kaijage SF, Michael K. Deep reinforcement learning based handover management for millimeter wave communication. *Int J Adv Comput Sci Appl (IJACSA).* 2021;12(2):784–91. <https://doi.org/10.14569/IJACSA.2021.0120298>.
109. Su Y, Lu X, Zhao Y, Huang L, Du X. Cooperative communications with relay selection based on deep reinforcement learning in wireless sensor networks. *IEEE Sens J.* 2019;19(20):9561–9. <https://doi.org/10.1109/JSEN.2019.2925719>.
110. Yang Z, Liu Y, Chen Y. Distributed reinforcement learning for NOMA-enabled mobile edge computing. In: *Proc. IEEE International Conference on Communications Workshops (ICC Workshops)*, Dublin, Ireland, 7–11 June 2020; pp. 1–6. <https://doi.org/10.1109/ICCWorkshops49005.2020.9145457>.
111. Sliwa B, Wietfeld, CA (2020) A reinforcement learning approach for efficient opportunistic vehicle-to-cloud data transfer. In: *Proc. IEEE Wireless Communications and Networking Conference (WCNC)*, Seoul, Korea, 25–28 May 2020; pp. 1–8. <https://doi.org/10.1109/WCNC45663.2020.9120681>.
112. Ning W, Huang X, Yang K, Wu F, Leng S. Reinforcement learning enabled cooperative spectrum sensing in cognitive radio networks. *J Commun Netw.* 2020;22(1):12–22. <https://doi.org/10.1109/JCN.2019.000052>.
113. Nasir YS, Guo D. Multi-agent deep reinforcement learning for dynamic power allocation in wireless networks. *IEEE J Sel Areas Commun.* 2019;37(10):2239–50. <https://doi.org/10.1109/JSAC.2019.2933973>.
114. Xu X, Tao M, Shen C. Collaborative multi-agent multi-armed bandit learning for small-cell caching. *IEEE Trans Wirel Commun.* 2020;19(4):2570–85. <https://doi.org/10.1109/TWC.2020.2966599>.
115. Hua Y, Li R, Zhao Z, Chen X, Zhang H. GAN-powered deep distributional reinforcement learning for resource management in network slicing. *IEEE J Sel Areas Commun.* 2020;38(2):334–49. <https://doi.org/10.1109/JSAC.2019.2959185>.
116. Sharma MK, Zappone A, Assaad M, Debbah M, Vassilaras S. Distributed power control for large energy harvesting networks: a multi-agent deep reinforcement learning approach. *IEEE Trans Cognit Commun Netw.* 2019;5(4):1140–54. <https://doi.org/10.1109/TCCN.2019.2949589>.
117. Zhong C, Gursoy MC, Velipasalar S. Deep reinforcement learning-based edge caching in wireless networks. *IEEE Trans Cognit Commun Netw.* 2020;6(1):48–61. <https://doi.org/10.1109/TCCN.2020.2968326>.
118. Kong J, Wu ZY, Ismail M, Serpedin E, Qaraqe KA. Q-learning based two-timescale power allocation for multi-homing hybrid RF/VLC networks. *IEEE Wirel Commun Lett.* 2020;9(4):443–7. <https://doi.org/10.1109/LWC.2019.2958121>.
119. Vu HV, Farzanullah M, Liu Z, Nguyen DHN, Morawski R, Le-Ngoc T. Multi-Agent Reinforcement Learning for Channel Assignment and Power Allocation in Platoon-Based C-V2X Systems. In: *Proc. IEEE 95th Vehicular Technology Conference: (VTC2022-Spring)*, Finland, 19–22 June 2022, pp. 1–5. <https://doi.org/10.1109/VTC2022-Spring54318.2022.9860518>.
120. Yazdinejad A, Parizi RM, Dehghantanha A, Choo KKR. Blockchain-enabled authentication handover with efficient privacy protection in SDN-based 5G networks. *IEEE Trans Netw Sci Eng.* 2021;8(2):1120–32. <https://doi.org/10.1109/TNSE.2019.2937481>.
121. Sana M, De Domenico A, Strinati EC, Clemente A. Multi-agent deep reinforcement learning for distributed handover management in dense mmWave networks. In: *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Barcelona, Spain, 4–8 May 2020; pp. 8976–8980. <https://doi.org/10.1109/ICASSP40776.2020.9052936>.
122. Ciftler BS, Abdallah M, Alwarafy A, Hamdi M. DQN-based multi-user power allocation for hybrid RF/VLC networks. In: *Proc. ICC 2021-IEEE International Conference on Communications*, Montreal, QC, Canada, 14–23 June 2021; pp. 1–6. <https://doi.org/10.1109/ICC42927.2021.9500564>.
123. Ye H, Li GY, Juang BHF. Deep reinforcement learning based resource allocation for V2V communications. *IEEE Trans Veh Technol.* 2019;68(4):3163–73. <https://doi.org/10.1109/TVT.2019.2897134>.
124. Zhang P, Wu M, Zhu X. Research on network fault detection and diagnosis based on deep Q learning. In: Wu Q, Zhao K, Ding X, editors. *Wireless and Satellite Systems WiSATS 2020. Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering*, vol. 358. Cham: Springer; 2021. https://doi.org/10.1007/978-3-030-69072-4_43.
125. What is deep learning ? in IBM cloud education page, may, 2020. <https://www.ibm.com/cloud/learn/deep-learning>.
126. Zappone A, Renzo MD, Debbah M. Wireless networks design in the era of deep learning: model-based, AI based, or both? *IEEE Trans Commun.* 2019;67(10):7331–76. <https://doi.org/10.1109/TCOMM.2019.2924010>.
127. He Y, Yu FR, Zhao N, Leung VC, Yin H. Software-defined networks with mobile edge computing and caching for smart cities: a big data deep reinforcement learning approach. *IEEE Commun Mag.* 2017;55(12):31–7. <https://doi.org/10.1109/MCOM.2017.1700246>.
128. Sutton RS, Barto AG. Reinforcement learning: an introduction. MIT press; 2018. ISBN: 9780262039246. <https://mitpress.mit.edu/9780262039246/reinforcement-learning/>.
129. Giupponi L, Perez-Neira AI. Fuzzy-based spectrum handoff in cognitive radio networks. In: *Proc. 3rd International Conference on Cognitive Radio Oriented Wireless Networks and Communications (Crown Com 2008)* 15–17 May 2008, Singapore, 2008; pp. 1–6. <https://doi.org/10.1109/CROWNCOM.2008.4562535>.

130. Tumuluru VK, Wang P, Niyato D. A neural network based spectrum prediction scheme for cognitive radio. In: Proc. 2010 IEEE International Conference on Communications (IEEE ICC), 23–27 May, Cape Town, 2010; pp. 1–5. <https://doi.org/10.1109/ICC.2010.5502348>
131. Tsagkaris K, Katidiotis A, Demestichas P. Neural network-based learning schemes for cognitive radio systems. *Comput Commun.* 2008;31(14):3394–404. <https://doi.org/10.1016/j.comcom.2008.05.040>.
132. Wang L, Han D, Zhang M, Wang D, Zhang Z. Deep reinforcement learning-based adaptive handover mechanism for VLC in a hybrid 6G network architecture. *IEEE Access.* 2021;9:87241–50. <https://doi.org/10.1109/ACCESS.2021.3089521>.
133. Li T, Liu W, Zeng Z, Xiong N. DRLR:(2022) A deep reinforcement learning based recruitment scheme for massive data collections in 6G-based IoT networks. *IEEE Internet Things J.* 2022;9(16):1495–609. <https://doi.org/10.1109/JIOT.2021.3067904>.
134. Shao X, Chen X, Qiang Y, Zhong C, Zhang Z. Feature-aided adaptive-tuning deep learning for massive device detection. *IEEE J Sel Areas Commun.* 2021;39(7):1899–914. <https://doi.org/10.1109/JSAC.2021.3078500>.
135. Xu H, Wu J, Li J, Lin X. Deep-reinforcement-learning-based cybertwin architecture for 6G IIoT: an integrated design of control, communication, and computing. *IEEE Internet of Things J.* 2021;8(22):16337–48. <https://doi.org/10.1109/JIOT.2021.3098441>.
136. Sami H, Otrouk H, Bentahar J, Mourad A. AI-based resource provisioning of IoE services in 6G: a deep reinforcement learning approach. *IEEE Trans Netw Serv Manag.* 2021;18(3):3527–40. <https://doi.org/10.1109/TNSM.2021.3066625>.
137. Mei J, Wang X, Zheng K, Boudreau G, Sediq AB, Abou-Zeid H. Intelligent radio access network slicing for service provisioning in 6G: a hierarchical deep reinforcement learning approach. *IEEE Trans Commun.* 2021;69(3):6063–78. <https://doi.org/10.1109/TCOMM.2021.3090423>.
138. Roy C, Saha R, Misra S, Dev K. Micro-safe: microservices-and deep learning-based safety-as-a-service architecture for 6G-enabled intelligent transportation system. *IEEE Trans Intell Transp Syst.* 2022;23(7):9765–74. <https://doi.org/10.1109/TITS.2021.3110725>.
139. Kalogiros C, Muschamp P, Caruso G, Hallingby HK, Darzanos G, Gavras A. Capabilities of business and operational support systems for pre-commercial 5G test beds. *IEEE Commun Mag.* 2021;59(12):58–64. <https://doi.org/10.1109/MCOM.003.2001059>.
140. Yifei S, Liu C. Applications of Artificial Intelligence in 5G, ZTE Technologies, no.4, issue 177, Date of release: 20.8.2018. 2018. <https://www.zte.com.cn/global/about/magazine/zte-technologies/2018/4-en/Special-Topic/4.html>.
141. Willars P, Wittenmark E, Ronkainen H, Östberg C, Johansson I, Strand J, Lédl P, Schnieders D. (2021) Enabling time-critical applications over 5G with rate adaptation, Ericsson - Deutsche Telekom White Paper, BNEW-21:025455, pp. 1–21, May 2021. Available: <https://www.ericsson.com/49bc82/assets/local/reports-papers/white-papers/26052021-enabling-time-critical-applications-over-5g-with-rate-adaptation-whitepaper.pdf>
142. Onyema EM, Almuzaini KK, Onu FU, Verma D, Gregory US, Puttaramaiah M, Afriyie RK. Prospects and challenges of using machine learning for academic forecasting. *Comput Intell Neurosci.* 2022. <https://doi.org/10.1155/2022/5624475>. (Article ID 5624475).
143. Geurts P, Irrthum A, Wehenkel L. Supervised learning with decision tree-based methods in computational and systems biology. *Mol Biol Syst.* 2009;5(12):1593–605. <https://doi.org/10.1039/b907946g>.
144. Mu J, Jing X, Zhang Y, Gong Y, Zhang R, Zhang F. Machine learning-based 5G RAN slicing for broadcasting services. *IEEE Trans Broadcast.* 2022;68(2):295–304. <https://doi.org/10.1109/TBC.2021.3122353>.
145. Elsayed M, Erol-Kantarci M. AI-enabled future wireless networks: challenges, opportunities, and open issues. *IEEE Veh Technol Mag.* 2019;14(3):70–7. <https://doi.org/10.1109/MVT.2019.2919236>.
146. Liu Y, et al. Reconfigurable intelligent surfaces: Principles and opportunities. *IEEE Commun Surv Tutor.* 2021;23(3):1546–77. <https://doi.org/10.1109/COMST.2021.3077737>.
147. Tang F, Mao B, Kawamoto Y, Kato N. Survey on machine learning for intelligent end-to-end communication towards 6G: from network access, routing to traffic control and streaming adaption. *IEEE Commun Surv Tutor.* 2021;23(3):1578–98. <https://doi.org/10.1109/COMST.2021.3073009>.
148. Hu YC, Patel M, Sabella D, Sprecher N, Young V. Mobile Edge Computing - A key technology towards 5G, Whitepaper of European Telecommunications Standards Institute (ETSI), France, September 2015, ISBN No. 979-10-92620-08-5, 2015; pp. 4–14. https://www.etsi.org/images/files/etsiwhitepapers/etsi_wp11_mec_a_key_technology_towards_5g.pdf.
149. Dong C, Shen Y, Qu Y, Wang K, Zheng J, Wu Q, Wu F. UAVs as an intelligent service: boosting edge intelligence for air-ground integrated networks. *IEEE Netw.* 2021;35(4):167–75. <https://doi.org/10.1109/MNET.011.2000651>.
150. Letaief KB, Shi Y, Lu J, Lu J. Edge artificial intelligence for 6G: vision, enabling technologies, and applications. *IEEE J Sel Areas Commun.* 2022;40(1):5–36. <https://doi.org/10.1109/JSAC.2021.3126076>.
151. Edge Computing Resources- Intel. <https://www.intel.com/content/www/us/en/edge-computing/what-is-edge-computing.html>.
152. Ren Z, Li X, Jiang Q, Wang Y, Ma J, Miao C. Network slicing in 6G: an authentication framework for unattended terminals. *IEEE Netw.* 2022. <https://doi.org/10.1109/MNET.112.2100738>.
153. Abella A, Chan C. 5G using Edge Computing Offers Low Latency, and Cost effective Innovation Opportunities, White paper of Intel, pp. 1–4. <https://www.intel.com/content/dam/www/public/us/en/documents/white-papers/5g-using-edge-computing-whitepaper.pdf>.
154. Ziegler V, Schneider P, Viswanathan H, Montag M, Kanugovi S, Rezaei A. Security and Trust in the 6G era, Whitepaper of Nokia Bell Labs Institute, August, 2021. pp. 1–22. <https://www.bell-labs.com/institute/white-papers/security-and-trust-6g-era/#gref>.
155. Okuno M, Shimada K, Fujiwara R, Ikeda N. Research and Development of DX Solutions for Social Infrastructure Using 5G/6G and AR/VR. *Hitachi Rev.* 2021;70(4): 75–80. https://www.hitachi.com/rev/archive/2021/r2021_04/04b03/index.html.
156. Kairouz EBP, McMahan HB. Advances and open problems in federated learning. *Found Trends Mach Learn.* 2021;14(1):1–210. <https://doi.org/10.1561/22000000083>.
157. Yang Q, Liu Y, Chen T, Tong Y. Federated machine learning: Concept and applications. *ACM Trans Intell Syst Technol.* 2019;10(2):1–19. <https://doi.org/10.1145/3298981>.
158. IEEE Std 3652.1-2020. IEEE guide for architectural framework and application of federated machine learning, 2020. https://standards.ieee.org/standard/3652_1-2020.html.
159. Park J, et al. Communication-efficient and distributed learning over wireless networks: principles and applications. *Proc IEEE.* 2021;109(5):796–819. <https://doi.org/10.1109/JPROC.2021.3055679>.
160. Gupta O, Raskar R. Distributed learning of deep neural network over multiple agents. *J Netw Comput Appl.* 2018;116:1–8. <https://doi.org/10.1016/j.jnca.2018.05.003>.
161. Chen T, Zhang K, Giannakis GB, Basar T. Communication-efficient policy gradient methods for distributed reinforcement

- learning. *IEEE Trans Control Netw Syst.* 2022;9(2):917–29. <https://doi.org/10.1109/TCNS.2021.3078100>.
162. Zhang K, Yang Z, Liu H, Zhang T, Basar T. Fully decentralized multi-agent reinforcement learning with networked agents. In *Proc. Int. Conf. Mach. Learn. (ICML)*, 2018; pp. 5872–5881. <http://proceedings.mlr.press/v80/zhang18n/zhang18n.pdf>.
163. Sun Y, Shi W, Huang X, Zhou S, Niu Z. Edge learning with timeliness constraints: challenges and solutions. *IEEE Commun Mag.* 2020;58(12):27–33. <https://doi.org/10.1109/MCOM.001.2000382>.
164. Yang K, Jiang T, Shi Y, Ding Z. Federated learning via over-the-air computation. *IEEE Trans Wirel Commun.* 2020;19(3):2022–35. <https://doi.org/10.1109/TWC.2019.2961673>.
165. Xu C, Liu S, Yang Z, Huang Y, Wong KK. Learning rate optimization for federated learning exploiting over-the-air computation. *IEEE J Sel Areas Commun.* 2021;39(12):3742–56. <https://doi.org/10.1109/JSAC.2021.3118402>.
166. Shi W, Zhou S, Niu Z, Jiang M, Geng L. Joint device scheduling and resource allocation for latency constrained wireless federated learning. *IEEE Trans Wirel Commun.* 2021;20(1):453–67. <https://doi.org/10.1109/TWC.2020.3025446>.
167. Amiri MM, Gunduz D, Kulkarni SR, Poor HV. Convergence of update aware device scheduling for federated learning at the wireless edge. *IEEE Trans Wirel Commun.* 2021;20(6):3643–58. <https://doi.org/10.1109/TWC.2021.3052681>.
168. Zheng S, Shen C, Chen X. Design and analysis of uplink and downlink communications for federated learning. *IEEE J Sel Areas Commun.* 2021;39(7):2150–67. <https://doi.org/10.1109/JSAC.2020.3041388>.
169. Park J, Samarakoon S, Shiri H, Abdel-Aziz MK, Nishio T, Elgabri A, et al. Wireless network intelligence at the edge. 2019. <https://doi.org/10.48550/arXiv.1812.02858>.
170. Wang X, Han Y, Wang C, Zhao Q, Chen X, Chen M. In-edge AI: intelligentizing mobile edge computing, caching and communication by federated learning. *IEEE Netw.* 2019;33(5):156–65. <https://doi.org/10.1109/MNET.2019.1800286>.
171. Lim WYB, Luong NC, Hoang DT, Jiao Y, Liang YC, Yang Q, et al. Federated learning in mobile edge networks: a comprehensive survey. *IEEE Commun Surv Tutor.* 2020;22(3):2031–63. <https://doi.org/10.1109/COMST.2020.2986024>.
172. Park J, Samarakoon S, Bennis M, Debbah M. Wireless network intelligence at the edge. In: *Proc. IEEE*, 2019; 107(11), pp. 2204–2239. <https://doi.org/10.1109/JPROC.2019.2941458>.
173. Li T, Sahu AK, Talwalkar A, Smith V. Federated learning: challenges, methods, and future directions. *IEEE Signal Process Mag.* 2020;37(3):50–60. <https://doi.org/10.1109/MSP.2020.2975749>.
174. Aledhari M, Razzak R, Parizi RM, Saeed F. Federated learning: a survey on enabling technologies, protocols, and applications. *IEEE Access.* 2020;8:140699–725. <https://doi.org/10.1109/ACCESS.2020.3013541>.
175. Murshed MG, Murphy C, Hou D, Khan N, Ananthanarayanan G, Hussain F. Machine learning at the network edge: a survey. *ACM Comput Surv.* 2022;54(8):1–37. <https://doi.org/10.1145/3469029>. (Article no. 170).
176. Yang Z, Chen M, Wonga K-K, Poor HV, Cui S. 6G Requirements, vision, and enabling technologies-review federated learning for 6G: applications. *Challenges Opportunities Eng.* 2022;8:33–41. <https://doi.org/10.1016/j.eng.2021.12.002>.
177. Ansi AA, Al-Ansi AM, Muthanna A, Elgendy IA, Koucheryavy A. Survey on intelligence edge computing in 6G: characteristics. Challe Potential Use Cases Market Drivers *Future Internet.* 2021;13(118):1–23. <https://doi.org/10.3390/fi13050118>.
178. Liu Y, Yuan X, Xiong Z, Kang J, Wang X, Niyato D. Federated learning for 6G communications: challenges, methods, and future directions. *China Commun.* 2020;17(9):105–18. <https://doi.org/10.23919/JCC.2020.09.009>.
179. Mikki S. The Shannon information capacity of an arbitrary radiating surface: an electromagnetic approach. *IEEE Trans Antennas Propag.* 2023;71(3):2556–70. <https://doi.org/10.1109/TAP.2023.3234164>.
180. Hakeem SAA, Hussein HH, Kim HW. Vision and research directions of 6G technologies and applications. *J King Saud Univ Comput Inform Sci.* 2022;34(6):2419–42. <https://doi.org/10.1016/j.jksuci.2022.03.019>.
181. Ssengonzi C, Kogeda OP, Thomas OO. A survey of deep reinforcement learning application in 5G and beyond network slicing and virtualization. *Array.* 2022;14(1–27):2022. <https://doi.org/10.1016/j.array.2022.100142>.
182. Gong Y, Yao H, Wang J, Li M, Guo S. Edge intelligence-driven joint offloading and resource allocation for future 6G Industrial Internet of Things. *IEEE Trans Netw Sci Eng.* 2024. <https://doi.org/10.1109/TNSE.2022.3141728>.
183. Berggren V, Dey K, Jeong J, Guldogan B. Bringing reinforcement learning solutions to action in telecom networks, A report—March 17, 2022, Ericsson. 2022. <https://www.ericsson.com/en/blog/2022/3/reinforcement-learning-solutions>.
184. Shen Y, Shi Y, Zhang J, Letaief KB. Graph neural networks for scalable radio resource management: Architecture design and theoretical analysis. *IEEE J Sel Areas Commun.* 2021;39(1):101–15. <https://doi.org/10.1109/JSAC.2020.3036965>.
185. Giannakas T, Spyropoulos T, Smid O. Fast and accurate edge resource scaling for 5G/6G networks with distributed deep neural networks. In: *Proc IEEE 23rd International Symposium on a World of Wireless, Mobile and Multimedia Networks (WoW-MoM)*, Belfast, UK, 14–17th June, 2022; pp. 100–109. <https://doi.org/10.1109/WoWMoM54355.2022.00021>.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.